

Sixteen Scientists on One Board

A society of piece-agents as a chess player, in which coordination is quorum sensing, strategy is a coalition formula, and a ply is a join

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Abstract

This note specifies a chess player whose *player* is not an evaluator but a society: each piece of the moving colour is an agent in its own right, and the sixteen agents coordinate to emit one move per turn. The board is likewise an agent. The setting is cost-accounted ρ as implemented on the `cost-accounted-rho` branch of `f1r3node-rust`, with the generated spatial-behavioural logic of the mortal-scientist programme, and we assume at development time an implementation of coalition logic augmented with context-labelled modalities. We argue five structural points and then give a spec. First, the board needs three name families per square, not one: an occupancy cell, a move interface, and an attack-marker family, because perception must not be able to act and because the attack map is the one expensive object all sixteen agents share. Second, a chess ply is not a single rewrite but an N -ary *join*, and the join arity is exactly the move taxonomy: two for a quiet move, two with occupant termination for a capture, three for en passant, four for castling. This is what preserves the generated modality $\langle K_j \rangle$ and the bijection between redex positions and legal moves. Third, the coordination problem is the honeybee problem, and its solution is forced by the same altitude argument: consensus over sixteen pieces is a global universal with no presented witness, quorum at a candidate move is a local $\oplus \otimes$ over one open, so the society must quorum-sense. Fourth — and this is the load-bearing observation — in ρ a quorum *is* a join arity and one linear advocacy token per piece per round is the polity's franchise, so the two-quorum theorem becomes an arithmetic fact about sends rather than a simulation result: with sixteen pieces and threshold $k \geq 9$ a split is structurally impossible. Fifth, legality is a genuine universal and therefore belongs to the world, not to the society; pins and checks are deposited structure and pin-awareness is perception. Sixth, the context-labelled modalities are not an evaluation device standing beside search — they *are* the search, with modal depth as ply depth, so lookahead is reflection: the checker quotes the deposit, manufactures a private shadow board, and drives it. This makes the price scale $\kappa(d) = 2^d$ derived rather than chosen, makes search depth endogenous, per-agent and funded by material, recasts the coalition rung in its forcing form with the opponent as the complement coalition, and puts pruning on the selection side of the same K/J split the commit gate sits on, so that a Viterbi witness at depth d is a principal variation. Seventh, what governs an evaluation term in this setting is not expressibility but price, and price turns on what the world deposits: a king-zone walk in the settling turns king safety into a depth-one read, so the short ladder is a short ladder and not a weak logic, and the real open question relocates from the hypothesis language to the hypothesis *search*. We accordingly decline to predict that this player will be weak for any structural reason. It is not that it cannot search; the one remaining candidate — that a quorum aggregates by plurality where an engine aggregates by maximum — is offset by a coverage gain from sixteen heterogeneous searchers that no draft has netted, and a measurement is specified to settle it. We then observe that capture is predation in the existing sense, so material advantage is metabolic advantage and the classical piece values become something the ecology might derive rather than receive.

The implementation spec that follows fixes the namespace discipline, the turn protocol, the formula ladder, the treasury — a fixed reserve disbursed on a schedule keyed to captured material, so that the society is deepest exactly where it is structurally split-free — the cost schedule, the requirements placed on the assumed coalition-logic checker, a six phase delivery plan, and eight measurements, of which the sharpest asks whether the noughts-and-crosses finding that non-modal rungs deliver ninety-two percent of yield for thirty-four percent of cost is a fact about games or a fact about noughts.

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1 Introduction

A conventional chess program is a single agent. It holds a position, generates moves, evaluates leaves, and returns an argmax. Everything about its architecture follows from that: the evaluation function is global, the search is centralised, and the pieces are data.

The proposal this note specifies inverts that. Each piece of the moving colour is an agent with its own budget, its own hypothesis about the position, and its own capacity to act. The board is an agent too. Sixteen agents want to move and exactly one move commits, so the architecture is obliged to contain something the conventional design never needs: a mechanism that turns sixteen concurrent desires into one commitment, without a chairman.

That obligation is the reason to build the thing. It is not a distributed implementation of a familiar algorithm. It is the smallest classical domain in which the constructions

of the mortal-scientist programme — cost-accounted computation, the generated spatial-behavioural logic, the assay in guard position, predation as the sole route to budget, and coalition logic read as a hypothesis language about populations — all have non-trivial content at the same time, and in which a century of external calibration is available to say whether they are worth anything.

1.1 What is being claimed, and what is not

What is claimed is that the society is a better instrument than a conventional engine for a specific question the programme has already asked and answered only in miniature, which is whether the yield of a hypothesis is worth its price; and that chess is the smallest board on which the answer could plausibly differ from the answer noughts-and-crosses gave.

Earlier drafts of this note surrounded that claim with an apology, and the apology kept being wrong in a way worth recording, because each version located the limit somewhere it was not.

Yield and strength are different quantities. They must be, or the programme has nothing to measure. The noughts result *is* that they come apart: the best marginal purchase on that ladder was not the best policy, and χ — the exact characteristic formula, the truth — was dominated on cost by a cruder rung. So a weak player and a handsome yield are perfectly compatible, and no statement about this player's strength is an answer, positive or negative, to the question the note is built around. That ought to have been said at the outset and was not.

It is not that the society cannot search. A first pass at this architecture suggests exactly that — sixteen budget-bounded agents paying a rendezvous per candidate, no central evaluator, no global depth parameter, in a game that rewards depth above almost everything else — and an earlier draft wrote the conclusion out. The suggestion ignores what the context-labelled modalities are for. They do the lookahead, and §9 works it out: modal depth is ply depth, evaluating $\langle\langle K_m \rangle\rangle\varphi$ means exhibiting the rewrite, and a society with the shared search memory of §9.5 is recognisably a loosely-coupled parallel search over a common record, so its coordination overhead is a constant factor per turn rather than an exponent.

Nor is it the logic. A second draft retreated to the position that the evaluation at the leaves is coarse and that this is forced, on the grounds that a rung must be a formula of the generated logic. That confuses a short ladder with a weak logic. The logic has ν , graded separation over described namespaces, the name predicate, and context-labelled modalities indexed by redex position, and under reflection it reaches further still. King safety and pawn structure are expressible, and §12.4 now carries both. What governs is not expressibility but *price*, and price turns on what the world deposits: a king-zone family costs one more walk in the settling and turns king safety into a depth-one read. The coarse ladder was a short ladder, and it is on the chosen list.

What is left. One item, and it is asserted here rather than established. A conventional engine takes the maximum over its candidates; a quorum takes a plurality above a threshold, so there is a gap at the commit between what the society plays and the best move any agent advocated. Against it runs a coverage gain that the same architecture supplies: sixteen agents at heterogeneous depths over different candidate sets search a wider tree than any one of them could. Whether the gap exceeds the gain is an empirical question which

M10 is specified to net out, and the note does not know the answer. §14 keeps the rest of the ledger, all of it chosen: the round budget, the candidate cap, the length of the ladder, and the treasury schedule of §10.1.

1.2 The spectrum, and where the interesting structure is

The design space is usually presented as a spectrum. At one extreme f1r3lang coordinates sixteen sessions with different language models, so a piece-agent is a thin wrapper around a model session. At the other extreme each agent is f1r3lang logic throughout. Section 11 argues that the extremes are not really two points on one dial. They are two answers to a single question — where a piece’s policy lives, and whether it is priced — and that the programme already has a construction that lets one have both without incoherence, namely the aperture. The recommendation is not a compromise between the extremes but a relocation: the model goes outside the ply loop, where it is good at the jobs it is good at, and inside only as a declared aperture in a stratified shard.

1.3 Organisation

Section 2 states what is imported from the existing programme and what is assumed to exist at development time. Section 3 argues for chess as the next worked example. Sections 4–10 are the analysis: two societies, the board, the ply as a join, the coordination problem, coalition logic, lookahead as reflection, and material as metabolism. Section 11 handles the model axis. Section 12 is the implementation spec and is the longest section. Section 13 lists the measurements. Section 14 is the honest accounting of what will go wrong. Section 16 lists open problems. Appendix A is the rho listing and Appendix B the formula catalogue.

2 What is imported, and what is assumed

2.1 From the mortal-scientist programme

The note stands on five constructions that already exist and are not re-derived here.

The cut and the three grades of access. A computation is ontologically isolated; the environment of a computation is other computations; and a learner reaches its environment by perception (reading deposited structure, priced by formula depth), interaction (rendezvous, priced by the communication) or reflection (quotation). The grades are separated in practice by a choice of namespace, not by a modality: the reason a threat marker in the noughts listing lives on its own name family is that were a line to signal a threat on a square’s move channel, the signal would rendezvous with the square’s receipt and the line would have *played the move*. Perception must not be able to act.

The generated logic. One primitive modality $\langle K_j \rangle$ per rewrite rule and choice of redex position, with j ranging over redex positions; structural connectives including separating conjunction; the name predicate $@\varphi$; graded separation over a *described namespace*, $\varphi \mid_k^N \psi$; greatest fixed point ν . There is no hidden-name quantifier and no need of one, because *new* is syntactic sugar: a name is a quoted process, no process contains the name that codes it, so a name fresh in P is manufactured by quoting any term properly containing P . Consequently every family of names used below is *described* by a predicate rather than enumerated, which is what makes graded separation over it available.

The assay. A hypothesis is not consulted and then acted on. It sits in guard position in a *where* clause, so acting *is* testing and confirmation *is* the rendezvous. The complementary guard pair yields three outcomes — confirm, refute, and a third that is empty, in which the budget was spent and nothing was learned. The third is ignorance, not refutation, and it is indistinguishable from the environment being faulty. That identification of the two bottoms is imported wholesale.

Conservation and predation. Nothing mints. The arena routes. On a win the arena hands the winner the loser’s metabolic name, so $\sigma + \kappa = \Theta$ throughout. An exogenous reward would need a referee with a mint and a referee with a mint destroys conservation.

Read versus drive. There are two routes to any predicate. Read it, at perception prices, if the environment has already computed it into legible structure; or drive it, priced by rendezvous. This is a claim about environments, not about learners, and it is statable only because the framework prices the grades separately and both parties have stacks. The noughts-and-crosses measurement found that the three non-modal rungs of its ladder delivered +0.444 of a total +0.481 gain — 92.3% of the yield — for 87.9 of 255.9 metered units, that is 34.3% of the cost.

2.2 From the consensus and quorum line

Coalition logic in Gabbay’s presentation, augmented with the behavioural connective of context-labelled modalities, with the graded reading developed in the quorum note: the coalition modality over a namespace N valued in a resolution semiring V ,

$$\langle E \rangle_N^V \psi = \bigoplus_{O \subseteq N \text{ open}} \bigotimes_{q \in O} \psi(q),$$

where opens are described sub-namespaces via the name predicate. Three results are imported. First, opens are not enumerated but described, so $\varphi \mid_k^N \psi$ applies to coalitions. Second, over $\mathbb{R}_{\geq 0}$ with $\otimes = \times$ a large coalition decays toward zero, so the correct quorum semantics is $\otimes = \min$, a coalition being as strong as its weakest member. Third, over the tropical and Viterbi algebras the attainment proposition holds on the nose, so the coalition modality is *attained* and the reader gets a witness coalition rather than a scalar.

The two-quorum theorem of the swarm note is imported in statement and re-proved in §7 in a form specific to this setting, where it is much stronger.

2.3 From the platform

Cost-accounted rho as it stands on `feature/cost-accounted-rho`. Two facts about the native integration bear on the spec. Metering is per-COMM at the reducer, keyed by the deploy envelope’s signature, and a signed term is *recognised* rather than lowered to a parking gate, so the normalised `Par` has the same COMM count as the unsigned program and there is no double-metering. In-program parking does not occur natively; an unfunded deploy is rejected at the acceptance gate rather than parked mid-program. The surface forms are unchanged: `s :: ()` mints, `{% P %}[s]` gates, `(*)` is joint authority, `::` stacks a budget, `-o` delegates, `# P` signs a section.

The live guard fragment is arithmetic and pattern only. Structural and modal guards are marked `//! ideal` in every listing in the repository, meaning they specify a check rather than perform one. Appendix A follows that convention exactly.

2.4 What is assumed to exist at development time

The instruction under which this note is written is that an implementation of coalition logic together with context-labelled modalities will be available. We therefore do not design around its absence, but we do state precisely what is being relied on, because a spec that assumes a component should say what the component must do. These are the requirements; they are collected again as an interface in §12.6.

Requirement 2.1 (Satisfaction, not synthesis). The checker must answer $P \models \varphi$. The consensus note’s term assignment runs the other way, $\vdash \varphi \triangleright P$, proofs as processes; the society needs the model-checking direction, because satisfaction is the guard side and a guard side is what a `where` clause is. Coalition logic here gives a piece a way to *ask* whether a coalition has formed, not a way to build one. This is the reason the whole synthesis-side gap of the consensus note — its missing proof system — is not on this project’s critical path.

Requirement 2.2 (Three-valued verdicts with a budget bottom). Verdicts are Kleene $\{f < b < t\}$ with b carrying both readings: the participant is faulty or stuck, and the observer could not afford to decide. The caller must not be able to distinguish them, and the spec below depends on that indistinguishability being visible in the measurements.

Requirement 2.3 (Semiring-parametric grading with attainment). The value algebra is a parameter. At minimum $\mathbb{B}$, $\mathbb{R}_{\geq 0}$, tropical $(\min, +)$ and Viterbi $(\max, \times)$ must be available. For the tropical and Viterbi instances the checker must return a *witness*: the coalition attaining the $\oplus$, not merely its value.

Requirement 2.4 (Indexed context-labelled modalities). $\langle K_j \rangle$ must be indexed by redex position, and the quantifiers $\exists j$ and $\forall j$ must range over redex positions of the presented term. An action-labelled logic will not do: every claim in the noughts world was the same action, and in chess every quiet move is the same rewrite rule. The distinctions that matter are positional.

Requirement 2.5 (Settled modality). A derived modality $\langle\langle K_j \rangle\rangle$ meaning “the rewrite at j fires, the board’s deposit computation settles, and then φ holds” must be available, with the settling shown terminating and confluent. In the noughts world this was Lemma 8; §5 restates the obligation for chess, where it is harder because ray scans are recursive.

Requirement 2.6 (Costed queries). Every query returns a metered cost alongside its verdict, charged to the caller’s stack, so that the ladder of §12.4 can be priced and the measurements of §13 can be taken. A checker that answers for free would make the entire experiment vacuous.

Requirement 2.7 (Namespace-indexed opens). The coalition modality must accept its namespace as a name predicate, so that “my two rooks” and “the pieces bearing on f7 ” are expressible without enumeration, and so that graded separation over the coalition namespace is available.

Requirement 2.8 (Caching by verdict). Repeated queries of the same formula against an unchanged deposit must be served from cache at the price of a lookup. Sixteen agents will ask overlapping questions; without caching the read-versus-drive measurement of §13 measures the checker’s implementation rather than the architecture.

3 Why chess, and why now

The programme’s existing worked example is noughts-and-crosses. Every number in it is exact over the 255,168 plays, which is its strength, and it is solved and tiny, which is its

weakness. Four things are true of chess and false of noughts, and each of them turns a stipulation of the earlier note into a question with an answer.

Material exists, so eating is non-trivial. In noughts nothing is captured; predation happens only at the arena, between games, when a loser's metabolic name is handed over. In chess predation happens *during* the game, at the granularity of a single ply, and the prey is a peer agent with its own stack. Conservation has to hold across that, and the accounting becomes interesting rather than ceremonial.

Coalitions are real, so coalition logic has content. Chess strategy is already coalition talk: the rook and bishop together force mate, these three pieces hold the kingside, that knight is doing two jobs. In noughts a coalition is a pair of threats and the vocabulary is overkill. In chess the coalition modality has something to quantify over.

The position is legible, so read-versus-drive is a real choice. The attack map of a chess position is expensive, shared, and recomputable. Whether the world deposits it or each agent drives it is a genuine design fork with a measurable cost difference. In noughts the analogous object is eight lines and the question does not bite.

The science cannot be closed, so the ecology does not die. The noughts note found that χ , the exact characteristic formula, is reached at nine unfoldings, and then found that this is fatal: a χ -monoculture draws every game, the harvest rate goes to zero, and the population starves on a finite wall. A solved science is a dead ecology and imperfection is load-bearing. In chess χ is not reachable at any budget the framework admits, so the ecology cannot die that way. We record this as the reason the ecological half of the programme becomes *observable* here for the first time.

Remark 3.1 (Chess keeps the ecology alive). The noughts result that a solved science is a dead ecology was obtained by exhibiting the solved case. It is a theorem about a closed science and its practical content — that real populations must be imperfect to eat — could not be observed in a world where perfection was affordable. Chess supplies a world where it is not. This does not make the earlier result stronger; it makes it testable in a regime where the alternative hypothesis, that the population converges on a single good policy and stops, is actually available.

Against those four, one thing is true of noughts and false of chess: exactness. Nothing in the chess project will be exact over the game tree. Every measurement in §13 is a sampled estimate against a fixed reference opponent, and the note should say so in its own results section rather than discovering it in review.

4 Two societies, one board

The proposal names two things as agents: the sixteen pieces and the board. It is worth being explicit that these are two *societies*, with different jobs, different budgets, and different logical characters.

Definition 4.1 (The world and the player). The *world* is the society of sixty-four square-agents together with the ray, attack, pin and legality machinery they support, plus the arena and the clock. The *player* is the society of at most sixteen piece-agents of one colour. The world holds no strategic stack and mints nothing; it deposits structure and refuses illegal joins. The player holds the budget and makes the decision.

This is the same split the noughts listing already makes, between `Board`, `Cells`, `Line` and `Scan` on one side and `Choose`, `Move` and `Revise` on the other, and it is the split that makes the question *who is billed* askable. The noughts note observed that putting threats in the term relocates the cost of the scan from the scientist's budget to the world's, and that the interesting question is who pays. In chess that question has a large answer, because the object in dispute is the attack map.

Remark 4.2 (The decomposition is a choice, and there are others). Agent-per-piece is not the only decomposition. Agent-per-square is the world. One could also try agent-per-ray (the sliding lines as agents, which is close to how the noughts `Line` contract behaves), or agent-per-plan, where an agent is a coalition and pieces are its resources. Agent-per-piece is chosen because it is the decomposition in which predation is definable: a piece can be captured and a ray cannot. The alternatives are worth naming because agent-per-plan is what a strong human player's introspection reports, and it is the natural second experiment.

5 The board

5.1 Why one name family per square is not enough

The proposal's encoding is

$$\prod_{1 \leq i, j \leq 8} @ (i, j) ! (\cdot)$$

with the payload `0` for unoccupied and `*x` for the fresh channel minted to interact with a particular piece. The occupancy half of this is right and the manufactured-name half is right — we return to both — but the encoding conflates two roles that the existing listings deliberately separate, and the reason they separate them is stated in the noughts source: perception must not be able to act, and that separation of access grades is enforced by choice of namespace.

In noughts the separation is cheap because perception is short-range: eight lines read three cells each. Chess has genuine long-range perception. A rook on `a1` must know what stands on `a8` without any intention of going there, and a bishop's diagonal passes over squares it will never occupy. If reading occupancy means consuming and re-emitting $@(i, j)!(v)$, then every perception is billed as an interaction, sixteen agents scanning concurrently serialise on every square, and the read-versus-drive distinction — which is one of the two things this project exists to measure — has been designed out of the system before the first line of code.

Definition 5.1 (Three name families per square). Let b be the board's channel and s a square index. The board maintains:

the occupancy cell

$occ(s) = @[*b, "occ", s]$, a resting send carrying either `0` or `*p` where p is the piece-agent's channel. This is the only family a ply consumes.

the attack family

$atk(s, c, p) = @[*b, "atk", s, c, p]$, standing offers deposited by the settling computation, one per attacker p of colour c bearing on s . Read-only in practice: no move protocol receives on it.

the constraint family

$pin(s)$, $chk(c)$, $ep(s)$, $cast(c, side)$, likewise deposited, carrying the derived facts a legality guard consults.

All three are manufactured names in the sense of the freshness-by-quotation proposition, not *new-bound*, so each family is describable by a name predicate and graded separation over it is available.

Proposition 5.2 (Perception must not act). *If attack markers are deposited on the occupancy family, then a scan that signals “ p bears on s ” can rendezvous with a move protocol receiving on $occ(s)$, and the scan performs a ply. Separating the families makes that failure unrepresentable rather than merely unlikely.*

Proof. Immediate, and it is the chess instance of the argument written into the noughts source for threat offers. The content is that the separation is structural: with distinct families there is no term in which a scan and a move protocol can rendezvous, so the property does not depend on any discipline the scan follows. $\square$

5.2 Who is billed for the attack map

The attack map is the expensive object in chess and it is shared by every agent in the player. There are two architectures.

World-billed.

After each ply the board runs a settling computation that walks every ray and deposits *atk* markers. The player’s agents then read markers at perception prices. One computation, sixteen readers.

Player-billed.

The board deposits only occupancy; each agent that needs to know whether a square is attacked drives the query itself, paying a rendezvous per square walked. Sixteen computations of one fact.

The framework’s own thesis says the first should win, and by a factor near the population size, because a predicate the environment has computed into legible structure is read at formula-depth prices rather than driven at rendezvous prices. But the framework does not say this is free: the world’s settling computation is metered too, and the noughts note’s remark on the point is precisely that the cost is *relocated*, not removed, and that the interesting question the framework can pose — because both parties have stacks — is who is billed. We therefore specify both and measure the difference (measurement M7, §13). This is the cheapest genuinely informative experiment in the whole project.

The same fork applies to *evaluation* terms and not only to legality and attack. King safety and pawn structure are classically computed per evaluated position; deposited by the settling walk they become depth-one reads (§12.4). So M7 is a larger experiment than it first appears: the question is not merely who pays for the attack map, but how much of an evaluation function the world can be made to compute once for everyone.

Remark 5.3 (Settling in chess is harder than in noughts). The noughts settling lemma had an easy proof: each of eight lines fires once, no line’s output enables another because offers go to names no line reads, so the computation terminates and is confluent. The chess settling walks rays, and a ray’s walk is recursive in the board’s occupancy: a slider’s ray terminates at the first occupied square. Termination is still clear — each ray step consumes one square of a strictly decreasing remaining ray — and confluence still holds for the same reason, that markers are deposited on a family nothing in the settling reads. But the obligation must be discharged explicitly, because $\langle\langle K_j \rangle\rangle$ depends on it and so does every modal rung of the ladder.

5.3 Legality is a universal, and it belongs to the world

Check, pin, and discovered check are constraints on the legality of *every* piece's move, and they are global: whether the bishop on c4 may move depends on whether doing so exposes the king. No individual piece can establish this locally and no quorum of pieces can establish it either, because an illegal move is not a move and a vote does not make it one.

Principle 5.4 (The legality universal is the world's obligation). *The one unavoidable $\forall$ in chess is placed on the world's budget, not the player's. The world refuses illegal joins, the way the arena in the noughts world holds no stack and only routes. The consequence is that the player's own hypothesis language stays existential and finitely confirmable, which is the regime in which the asymmetry proposition says witnesses are cheap.*

There is a second consequence, and it is the one that makes this satisfying rather than merely convenient. If the world deposits *pin(s)* markers — computed by walking outward from each king through the first own piece to the first enemy slider — then pin-awareness becomes perception. A piece does not reason about whether it is pinned; it reads whether a marker stands on its square. Likewise check is a marker, not a search. The single most classically “knowledge-heavy” part of chess legality becomes deposited structure, and the read-versus-drive thesis gets its best instance.

Remark 5.5 (What the world does not deposit). The world deposits facts about the *current* position: occupancy, attacks, pins, checks, en-passant and castling rights. It does not deposit *evaluated* facts about the consequences of moves, because those are the player's hypotheses and pricing them to the world would smuggle the evaluation function back into the architecture through the floor. It should be guarded in review: any proposal to have the board deposit “this move loses a piece” is a proposal to write a conventional engine and call it an ecology.

The line is between evaluated and *searched* facts, not between depth zero and depth greater than zero, and §9.5 draws it. A record that a position was driven to depth d and returned verdict v is history rather than policy, and it is shareable for the same reason the attack map is. An earlier draft of this note drew the line at depth instead, which had the effect of forbidding shared search memory without argument.

6 A ply is a join

6.1 What the noughts construction got, and why chess does not inherit it

The second draft of the game note was a correction: the context K had never been instantiated, and instantiating it as a *redex* context was what made $\langle K_s \rangle$ a generated primitive rather than a derived abbreviation. The payoff was a bijection. An empty square is a receipt offering to be claimed; a claim is one join consuming the mover's send and the stale mark; after it fires no receipt remains. So redex positions correspond to empty squares correspond to legal moves, and the branching factor of $\exists s$ is the legal move count.

None of that transfers to chess for free. A chess move touches two squares at minimum, vacating one and occupying another. A capture additionally ends a sibling agent's computational life. Castling touches four squares and two pieces. En passant has a destination distinct from the square of the captured pawn. Promotion replaces a piece's identity. If a ply is several rewrites then $\langle K_j \rangle$ indexes something finer than a move, the bijection fails, the modal rungs of the ladder index the wrong thing, and — worse operationally — two plies can interleave in the middle of each other.

6.2 The repair

The repair is available and it is already demonstrated in the platform. The cost-accounting demonstration uses an N -ary atomic join for a four-cell swap that conserves money and goods all-or-nothing in every interleaving, with the guard deciding before any leg fires. A chess ply is that shape exactly.

Definition 6.1 (A ply). A ply is a single N -ary join over the occupancy cells it touches, guarded by a `where` clause consulting the constraint family, which re-emits every consumed cell with its updated payload. Formally, for a move m touching squares $s_1, \dots, s_n$,

$$\text{for } (@v_1 \leftarrow \text{occ}(s_1) \ \& \ \dots \ \& \ @v_n \leftarrow \text{occ}(s_n) \text{ where } \text{Legal}(m, \vec{v})) \{ \dots \}$$

where the body re-emits $\text{occ}(s_1), \dots, \text{occ}(s_n)$ with the post-move payloads, terminates any captured piece-agent, and triggers the settling.

Proposition 6.2 (Move taxonomy is join arity). *Under Definition 6.1 the classification of chess moves by arity and payload transformation is exactly the classical move taxonomy:*

Move kind	Arity	Payload transformation
Quiet move	2	$(*p, \mathbf{0}) \mapsto (\mathbf{0}, *p)$
Capture	2	$(*p, *q) \mapsto (\mathbf{0}, *p), q \text{ terminated}$
En passant	3	$(*p, \mathbf{0}, *q) \mapsto (\mathbf{0}, *p, \mathbf{0}), q \text{ terminated}$
Castling	4	$(*k, \mathbf{0}, \mathbf{0}, *r) \mapsto (\mathbf{0}, *k, *r, \mathbf{0})$
Promotion	2	$(*p, \cdot) \mapsto (\mathbf{0}, *p'), p \text{ retired}, p' \text{ born}$

We state this as a proposition rather than a table because it has a consequence.

Corollary 6.3 ($\langle K_j \rangle$ survives). *With plies as joins, redex positions of the ply rule correspond bijectively to legal moves, so $\langle K_j \rangle$ remains a generated modality with j ranging over moves, $\exists j$ has the legal move count as branching factor, and every modal rung of the ladder indexes what it is supposed to index. The whole ladder apparatus of the game note transfers.*

Remark 6.4 (The alternative, and why not). One could let a ply be several rewrites and recover a composite via the settled modality, as the noughts world does for the board's own settling. The cost is that a ply is then no longer atomic in the term, so two agents' plies can interleave, and the guard that makes a move legal would have to be re-checked against a position that may have changed under it. Chess has a turn structure and it should be represented by atomicity rather than by convention. The join route also costs nothing: the platform has N -ary joins and the demonstration of a conserving four-cell swap already exists.

Remark 6.5 (Promotion is birth and it is not free). Promotion is the only move that creates an agent. Under conservation it cannot create a *budget*: the promoted agent's stack must come from somewhere, and the natural source is the pawn's own stack, transferred whole. That makes promotion a change of policy and capability at constant budget, which is the right reading, and it means the classical observation that a promoted queen is worth a queen is not automatic here — it is a prediction, since the promoted agent inherits a pawn's accumulated stack rather than a queen's.

7 The coordination problem

7.1 The shape of the problem

Sixteen agents each want to move; exactly one move commits. This is not a technical inconvenience, it is the whole architectural content of the proposal, and the programme already contains the analysis of it.

Observation 7.1 (The K/J split). A graded clause is a map from candidates to values: a *weighting* of the fibre, not a *section* of it. In the selection-monad reading, a clause is a quantifier $(X \rightarrow R) \rightarrow R$ and a resolver is a selection function $(X \rightarrow R) \rightarrow X$. The swarm exhibits the split as different behaviours performed by different bees: dance evaluation is the clause, piping and takeoff are the section, and no bee does both.

The piece society must exhibit the same split. A piece *evaluates* candidate moves; something else *commits*. A design in which each piece both scores and plays is a design with sixteen resolvers and no game.

7.2 Quorum, not consensus, and the argument is not a preference

The obvious architecture is a council: all sixteen pieces report scores, the scores are combined, the argmax commits. That architecture has a chairman, and a chairman is a global evaluator wearing a hat. The programme's altitude argument rules it out, and it does so by derivation rather than by taste.

Proposition 7.2 (The mechanism is forced). *Consensus at the council is a global $\forall$ over all sixteen advocates with no presented witness: finitely refutable, not finitely confirmable. Quorum at a candidate move is a local $\oplus \otimes$ over one open. No participant can evaluate the global universal within a bounded budget, so a society of budget-bounded participants must use the local formula.*

This is the quorum note's derivation of Seeley's empirical finding, restated for a designed rather than an evolved population. In the swarm it explains why quorum is sensed at the site rather than at the cluster. Here it says: pieces advocate for candidate *moves*, and a candidate that accumulates enough advocacy commits. Nobody counts the whole distribution.

7.3 The advocacy namespace, and why a candidate move is a name

Definition 7.3 (Advocacy). A candidate move m is a term; by reflection $@m$ is a name. The advocacy channel for m is $adv(m) = @[*pl, "adv", @m]$ where pl is the player's channel. A piece advocates for m by sending its advocacy token on $adv(m)$. An *open* in the coalition sense is the set of pieces that have advocated for a given m during the current round; it is described by a name predicate, not enumerated.

This is the herd note's corollary landing exactly: opens are described namespaces via $@\varphi$. It is also, we think, the most pleasing single consequence of working in a reflective calculus. In the swarm an open is a set of bees and the modeller has to say so. Here the site a piece advocates for *is* a name, manufactured by quoting the move, and the advocacy set is a sub-namespace of the player's namespace. Nothing has to be introduced to make coalition logic apply; the calculus already provides the points.

7.4 One token per piece per round: the society is a polity

Definition 7.4 (The franchise). At the start of each advocacy round the player issues to each surviving piece exactly one *linear* advocacy token, of signature V , into a located stack whose channel is manufactured from the piece's identity. A piece spends its token by advocating, or by inhibiting, or not at all. Tokens do not carry over between rounds.

This is the polity listing's pre-election phase transposed without change: a persistent claim on the roll metered to one linear token per epoch, a source keyed to identity and not a mint. The reason to adopt it is not tidiness. It is that it makes the following theorem an arithmetic fact about sends rather than a statistical regularity.

7.5 Quorum is a join arity

Definition 7.5 (The commit gate). The commit gate for candidate m at threshold k is the k -ary join

$$\text{for } (@a_1 \leftarrow \text{adv}(m) \ \& \ \cdots \ \& \ @a_k \leftarrow \text{adv}(m)) \ \{ \text{Commit}(m) \}$$

which fires exactly when k distinct advocacy sends stand on $\text{adv}(m)$, consuming all of them.

Theorem 7.6 (Two quorums need $2k$ tokens). *Let S be the number of surviving pieces, each holding exactly one linear advocacy token per round, and let k be the commit threshold. Then two distinct candidate moves cannot both fire their commit gates in the same round unless $S \geq 2k$. In particular with $S = 16$ and $k \geq 9$ a split is impossible.*

Proof. Each commit gate consumes k sends on its own channel. Tokens are linear, so a token spent on $\text{adv}(m)$ is not available on $\text{adv}(m')$ for $m' \neq m$, and each piece holds one. Two firings therefore consume $2k$ distinct tokens drawn from a pool of S . $\square$

Remark 7.7 (What this theorem is worth). In the swarm the corresponding statement is a structural impossibility result that a statistical model cannot state, and it was confirmed by simulation: splits identically zero at $S = 12, 16, 20, 24, 28, 29$ and first appearing at $S = 30 = 2k$ with $k = 15$. Here it needs no simulation, because linearity of the token and arity of the join do the counting. That is the difference between importing a result and re-deriving it in a calculus that already had the right primitives. It also means the designer has a free safety property: choose $k \geq \lceil S/2 \rceil$ and the society *cannot* commit two moves, for any policies whatsoever, with no cross-inhibition machinery at all.

7.6 The other failure mode, and what cross-inhibition is for

Safety comes free; liveness does not. With $k \geq 9$ and advocacy split eight against eight, no gate fires and the round deadlocks. The swarm literature's division of labour applies exactly: below $2k$ cross-inhibition prevents *deadlock*; above $2k$ it prevents *split*. Since the design fixes $k \geq S/2$ and buys no-split structurally, cross-inhibition in this setting has only the deadlock job, and it competes with a cheaper mechanism.

Definition 7.8 (Rounds, and the deposited distribution). A turn consists of up to R advocacy rounds. At the end of a round in which no gate fired, unconsumed advocacy sends remain standing on their $\text{adv}(m)$ channels and are *readable*. Fresh tokens are issued. A piece may therefore re-advocate having perceived, at perception prices, where the rest of the society stood.

This is the bees reading dance strength at the cluster: cheap perceptual price for the reversible decision (which candidate to back), expensive interaction price for the irreversible one (the commit). We take that as the default and treat cross-inhibition as a labelled variant, for two reasons. It is one more mechanism, and the swarm measurement recorded an unexpected cost — with cross inhibition on, the swarm chose the better site slightly *less* often, because inhibition suppresses correct advocacy too.

Definition 7.9 (The resolver of last resort). If R rounds elapse with no commit, the clock commits the candidate carrying the most standing advocacy, ties broken by a deterministic order on $@m$.

This is the one centralised element in the design and the note should not hide it. It is held by the clock, which is world and not player; it has no evaluation function and no preferences; and its firing rate is a measurement (M4, §13), because a society that reaches last resort often is a society whose quorum parameters are wrong.

7.7 The population shrinks, and it crosses the boundary during play

Proposition 7.10 (The endgame crossing). *S is monotonically non-increasing over a game, from 16 toward as few as one piece besides the king. If k is held fixed as an absolute number, then $S < k$ eventually holds and no gate can ever fire; if k is held fixed as a fraction αS with $\alpha > 1/2$, splits remain impossible throughout but the absolute number of advocates required falls. Either way the society crosses a regime boundary at a computable ply.*

Remark 7.11 (A prediction the bees could not supply). The swarm's S is fixed within an episode, so the two-quorum theorem there partitions *episodes*. Chess partitions the *inside* of a single episode: the same society passes through the split-possible regime and into the split-impossible one as material comes off. The framework therefore predicts that the character of the decision procedure must change at a computable point, with competing plans available in the middlegame and structurally unavailable in the endgame. That this matches ordinary chess intuition — middlegames are about competing plans, endgames about executing one — is not evidence, but it is the kind of coincidence that makes a measurement worth taking rather than a post-hoc story worth telling. M5 in §13 is that measurement, and it is the one we would report first. Corollary 10.8 sharpens it: with the treasury of §10.1 the society is at its richest exactly where it is structurally split-free, so the two mechanisms point the same way and the prediction acquires a direction as well as a location.

8 Coalition logic as the society's strategic vocabulary

8.1 The vocabulary is already chess's own

Chess commentary is coalition talk. The rook and bishop together force mate. These three pieces hold the kingside. That knight is overloaded, doing two jobs at once. That bishop is bad, belonging to no coalition. Coalition logic's $\langle E \rangle \varphi$, "coalition E can force φ ", is not a formalism imported and then applied; it is the native register of the domain.

This matters for the project because it means the society's hypotheses can be *about* coalitions without anyone having designed a coalition-formation mechanism. Pieces advocate; advocacy sets are opens; opens are coalitions; and the logic evaluates formulae over them.

8.2 Which algebra, and why the answer is not $\mathbb{R}_{\geq 0}$

Proposition 8.1 (Large coalitions must not decay). *Over $\mathbb{R}_{\geq 0}$ with $\otimes = \times$ and values in $[0, 1]$, a coalition's value $\bigotimes_{q \in O} \psi(q)$ is non-increasing in $|O|$ and strictly decreasing whenever any member's value is below one. Hence larger quorums score weaker, which inverts the intended semantics.*

Definition 8.2 (Quorum algebras). A quorum algebra takes $\otimes = \min$: a coalition is as strong as its weakest member. Over the tropical $(\min, +)$ and Viterbi $(\max, \times)$ algebras the attainment proposition holds on the nose, so $\oplus$ is realised by a definable selection function.

We note, because it is a pleasant piece of evidence rather than an argument, that $\otimes = \min$ is independently the correct chess semantics. A combination fails at its weakest participant. If three pieces bear on a mating square and one of them is pinned, the combination is worth what the pinned piece is worth, which is nothing; the other two do not compensate. Overloading is the same shape from the other side: a piece that belongs to two coalitions caps both at its own availability. The algebra the framework forces on general grounds is the algebra the domain would have chosen.

Corollary 8.3 (The player gets a witness). *Over the attained algebras the coalition modality returns not a scalar but the coalition realising it. The society therefore commits to a move together with the coalition that justifies it.*

Remark 8.4 (Explanation is structural, not decorative). Corollary 8.3 is the closest thing this project has to a by-product worth having on its own. An explanation of the form “this move, because these pieces, at this strength” is not generated after the fact by a separate narration module reading the engine's numbers. It is the witness the selection function returned. Whatever else is true of the resulting player, its explanations cannot be confabulated, because there is no separate faculty available to confabulate with.

8.3 The alternating rung, and the question this project exists to answer

Coalition logic's “can force” quantifies over the opponent's replies. It is therefore an alternating formula, a box under a diamond, and the game note's measurement of the corresponding rung was unkind: rung 4, the one genuinely alternating rung of the noughts ladder, bought +0.005 for 111 metered units and was the worst purchase on the ladder, while the three non-modal rungs delivered 92.3% of the yield for 34.3% of the cost.

We do not think this settles anything, and there are now two reasons rather than one. The first is the reason for the whole project, below. The second is that the comparison is not like for like: rung 4 in noughts is a two-ply search on a game with 5478 positions, and rung 4 in chess is a two-ply search on chess, so a ratio computed against it in one setting is not a ratio computed against the same instrument in the other. §9 is where this is worked out and M2 is restated accordingly.

Observation 8.5 (The yield result is a measurement on one game). Noughts-and-crosses has 5478 reachable positions, no material, no long-range interaction, and a branching factor that starts at nine and falls. Its non-modal predicates — a completing threat, a fork, centrality — are close to sufficient because the game is close to trivial. Whether the finding generalises is not a question the noughts data can answer, because the finding could be a fact about games or a fact about the smallness of that game.

Principle 8.6 (Chess is the discriminating instance). *The measurement that would most change the programme's view of itself is the ladder measurement (M1, M2) repeated on chess. If the non-modal rungs again deliver most of the yield for a third of the cost, the read-versus-drive thesis acquires*

its second and much harder instance and becomes a claim worth defending in general. If the modal and coalition rungs dominate in chess, then the noughts result is a result about noughts, and the programme has learned something it could not have learned from a game it could solve.

This is the strongest strategic argument for building the thing, and it should appear in the note's own abstract, which it does.

8.4 Graded separation and what a fork is in chess

The noughts fork was $\text{Threat}(p) \mid_0^N \text{Threat}(p)$: two threats sharing no square, graded at separation zero over the square-indexed namespace. Requiring it is what forced the board encoding, since two threats with a common gap must be excluded and therefore squares rather than lines must be the parands.

The chess transposition is direct and the namespace is the attack family.

Definition 8.7 (Fork, chess instance). Let N_{atk} be the namespace described by the name predicate

$$@[*b, "atk", \cdot, c, \cdot],$$

and write $\text{Bears}(p, s)$ for the structural predicate asserting a marker deposited by p on s . Then

$$\text{Fork}(p) := (\exists s. \text{Bears}(p, s) \wedge \text{Valuable}(s)) \mid_0^{N_{atk}} (\exists s'. \text{Bears}(p, s') \wedge \text{Valuable}(s')).$$

The same argument that forced squares as parands in noughts forces *squares* here too, not pieces: a knight bearing twice on one square is not a fork, and separation over a piece-indexed namespace would not exclude it.

8.5 The coalition rung

Definition 8.8 (The coalition rung). For a target square s and the player's piece namespace N_{pl} ,

$$\text{Bearing}(s) := \langle E \rangle_{N_{pl}}^V \lambda p. \text{Bears}(p, s) \otimes \text{Available}(p),$$

where $\text{Available}(p)$ is zero if p is pinned or already committed elsewhere. Over a quorum algebra this evaluates to the strength of the strongest available coalition bearing on s , attained at a witness.

Remark 8.9 (Superseded by the forcing form). Definition 8.8 reads the coalition modality statically, as a count of who bears on a square weighted by availability. That is not what $\langle E \rangle$ means, and §9.3 replaces it with the forcing form. The static version is kept here because its Available factor survives unchanged into the replacement and because the discussion immediately following is about that factor.

Available is where pins enter, and it is worth observing that it enters as a *factor* in the $\otimes$ rather than as a side condition. A pinned piece contributes $\perp$ to any coalition containing it, and under $\otimes = \min$ that zeroes the coalition. This is the correct behaviour and it was not designed in; it follows from the choice of algebra forced in §8.

Remark 8.10 (Brittleness, and the two dials). The herd note recorded a negative result that applies here without change: min over a large coalition is zeroed by one dissenter, so a crisp rung and a graded rung can return identical rows, and grading pays only if coalition *size* is relaxed as well. The chess instance is that Bearing over a fixed large coalition is brittle, and the design must therefore relax over subsets — which it does, since $\oplus$ ranges over opens, and the witness returned is the best available subset rather than the whole. We flag it because it was a surprise in the earlier work and would be a surprise again.

9 Lookahead is reflection

9.1 The modalities are the search

The naive reading of this architecture says that search is outside its capacity. Sixteen budget-bounded agents, a rendezvous price on every candidate examined, no central evaluator and no global depth parameter, in a game with a branching factor near thirty-five that rewards depth above almost everything else: the conclusion writes itself, and an earlier draft of this note wrote it. That draft treated the modal rungs as evaluation predicates which happen to be expensive, and then conceded separately that the society would be a perception player rather than a search player.

Both halves are wrong, and they are wrong in the same way. They ignore what the context-labelled modalities are for. The modalities are not an evaluation device standing alongside search. They *are* the search.

Observation 9.1 (Modal depth is ply depth). $\langle\langle K_m \rangle\rangle\varphi$ asserts that the ply at m fires and then φ holds: one ply of lookahead. Rung 4, $\exists m \forall m'. \langle\langle K_m \rangle\rangle[K_{m'}] \neg \text{Fork}(\text{them})$, is a two-ply minimax written as a formula, with $\exists$ the maximising player and $\forall$ the minimising one. And $\chi = \nu X. \forall m. [K_m](\neg \text{Lost} \wedge \exists m'. \langle\langle K_{m'} \rangle\rangle X)$ is full game-tree search written as a greatest fixed point. That χ was exact at nine unfoldings in the noughts world is not a curiosity about the formula; nine unfoldings *is* solving the game.

Observation 9.2 (The price scale was search cost all along). The schedule $\kappa(d) = 2^d$ per candidate examined, charged by the depth of the formula a rung installs, was introduced in the game note as a choice and varied for sensitivity. Under Observation 9.1 it is the branching cost of a d -ply search wearing a different name. §9.2 makes the identification exact and thereby turns a chosen constant into a derived one.

The correct claim is therefore not that the society cannot search. It is this.

Principle 9.3 (Search depth is endogenous, per-agent, and funded). *A conventional engine has one global depth parameter, set from outside. Here the depth an agent can reach is the rung it can pay for, so depth is heterogeneous across the society and varies within a single turn. An agent that has eaten can calculate further than one that has not.*

Proposition 9.4 (Material buys calculation, not merely perception). *Proposition 10.2 says that a side with more surviving material has more advocates and more aggregate stack. Under Principle 9.3 the stack purchases modal depth, so the conventional evaluation term “material” resolves into affordable lookahead and not only into affordable perception.*

That is much closer to what material does in chess than the earlier reading was, and it raises the stakes on Conjecture 10.3: a queen’s value would then be partly the depth its bearer can afford, which is a quantity the experiment measures directly rather than a quantity it has to infer.

9.2 The shadow board

There is an obligation to discharge before any of this is legitimate. Requirement 12.7 says the checker must never receive on the occupancy family, on the grounds that a checker which moves pieces is a pride that moves the zebra by looking at it. But $\langle\langle K_m \rangle\rangle\varphi$ requires the ply to fire. In a rewriting calculus, evaluating a modal formula means exhibiting the rewrite. As stated, the requirement forbids exactly what the modal ladder needs.

The resolution is not a patch. It is the calculus’s own primitive.

Definition 9.5 (Shadow board). Let b be the board and H a handle on its current deposit. A *shadow board* at depth d is the board $@[*chk, H, d, \iota]$, manufactured by quotation from the checker’s own channel, the deposit handle, the depth and a branch index ι . Its occupancy, attack and constraint families are the families of Definition 5.1 over that name. The checker may drive a shadow board without restriction, because it is the checker’s own term.

Requirement 9.6 (The checker may not move *the* board). This supersedes the earlier statement. The checker must not receive on $occ(s)$ for the live board b ; it peeks there, never consumes. It may receive freely on the occupancy family of any shadow board it has manufactured. Since shadow names are quotes of terms properly containing the checker, they are fresh in the live board by the freshness proposition, so the two families cannot collide and the separation is structural rather than disciplinary.

Principle 9.7 (Lookahead is reflection). *Imagining a position is quoting the world and driving the copy. The one primitive that makes this available is the quote, which the calculus has had in its syntax since the beginning.*

The choice-formulae note’s strongest single line is that Krivine realises dependent choice by adding a `quote` instruction to the machine, whereas the rho calculus has `quote` as a term former, so reflection is the calculus’s native choice primitive. Principle 9.7 gives `quote` a second job in the same programme. It also supplies a mechanism the scientist note lacked: that note’s mixed-environments section lists imagination alongside theory theft as something a scientist in a populated environment can do, and says what it is for without saying what it is. It is this.

Proposition 9.8 (The price scale is derived). *Evaluating a rung- d formula at one candidate instantiates a shadow board and drives it through d plies. If β is the number of continuations examined per ply, the cost is $\Theta(\beta^d)$ shadow plies, each of which is a join of bounded arity. Taking the noughts convention $\beta = 2$ gives $\kappa(d) = 2^d$ exactly.*

Remark 9.9 (What this settles, and what it opens). §12.5 says that every price in the schedule is a choice rather than a derivation, and requires a sensitivity analysis over three settings as an acceptance criterion. Proposition 9.8 removes the modal rows from that requirement: their prices are now consequences of the architecture and the only free parameter is β , which is itself measurable rather than stipulated. The non-modal rows remain choices, and the sensitivity analysis stands for them. What this opens is that β is exactly the quantity pruning attacks, which is §9.4.

9.3 Forcing, and the opponent as complement coalition

Definition 8.8 used the coalition modality statically: a count, weighted by availability, of who bears on a square. That is a defect, and naming it is worth more than the definition was.

$\langle E \rangle \varphi$ does not mean “ E has the resources for φ ”. It means E can *force* φ whatever the complement of E does. The complement, here, is the opponent’s pieces. So the coalition modality is not a static scoring device that sits beside adversarial lookahead; it is where adversarial lookahead belongs in this vocabulary, and it brings the opponent in as the complement of a coalition rather than as a second search agent the society would have to simulate explicitly.

Definition 9.10 (The coalition rung, forcing form). For a target s and the player’s piece namespace N_{pl} ,

$$\text{Forces}(s) := \langle E \rangle_{N_{pl}}^V \nu X. ([\text{Mate}(s)] \oplus \langle K \rangle X),$$

read: some available coalition of my pieces can force mate at s against any play by the complement. Available(p) enters as a factor in the $\otimes$ exactly as before, so a pinned member still zeroes its coalition under $\otimes = \min$.

Proposition 9.11 (Rung 4 is a special case). *Rung 4's $\forall m'. \langle\langle K_m \rangle\rangle [K_{m'}] \neg \text{Fork}(\text{them})$ is Forces at coalition N_{pl} , fixed-point depth one, and target predicate $\neg \text{Fork}(\text{them})$ rather than Mate. The two rungs therefore stand in a refinement relation rather than in competition, and the ladder is a chain at the top as well as at the bottom.*

This also repairs an asymmetry in the pricing. The earlier ladder put rung 6 above rung 4 at $\kappa = 16$ without rung 6 doing anything rung 4 did not. In the forcing form the ordering is honest: rung 4 is depth one, rung 6 is a ν over the run tree and therefore a depth engine, which is where the herd note's altitude argument puts compactness and strong fairness, and it is priced as such.

9.4 Pruning is a scheduler discipline

The alternation $\exists \forall \exists$ is fixed by the formula. Which continuations one examines, and in what order, is not; it is a choice of evaluation order over the fibre of candidates.

Observation 9.12 (The split, a fourth time). The formula is the quantifier, the search order is the selection function. So alpha-beta is not a different formula from minimax, it is a different resolver for the same clause, and it belongs on the J side of the K/J split exactly as the commit gate does.

Proposition 9.13 (The witness is a principal variation). *Over an attained algebra — tropical or Viterbi, per Definition 8.2 — the selection realising $\oplus$ at modal depth d returns not a value but the sequence of redex positions attaining it. That sequence is a line of play of length d .*

Corollary 9.14 (Soundness of pruning is attainment). *A pruning discipline is sound exactly when it does not disturb the attaining selection, that is, when the $\oplus$ remains attained at the same witness after candidates are skipped. The soundness condition for alpha-beta is therefore the attainment condition the note already imports, and not a separate argument.*

Proposition 9.13 is Corollary 8.3 one level up. At depth zero the witness is the coalition that justifies the move; at depth d it is the line that justifies it. The society therefore commits to a move together with both, and Remark 8.4 strengthens: an explanation of the form “this move, because these pieces, along this line” is the witness the selection function returned, not a narration assembled afterwards from scores.

9.5 Shared search memory

Remark 5.5 drew a line: the world deposits facts about the current position and not about the consequences of moves, because pricing consequences to the world would smuggle the evaluation function back in through the floor. That is right about *evaluated* facts and wrong about *searched* ones, and the distinction was invisible while lookahead was mis-classified.

Definition 9.15 (Searched versus evaluated). An *evaluated* fact is the verdict of a policy: this move is good, this square is weak. A *searched* fact is the record of a computation: the position with deposit-hash H was reached, driven to depth d , and returned verdict v with witness w at price c . The first is policy and belongs to the agent that holds it. The second is history and belongs to nobody.

Principle 9.16 (Read-versus-drive applies at every depth). *Sixteen agents doing independent depth- d lookahead over overlapping candidate sets duplicate an enormous amount of shadow driving. A searched fact deposited once is read by the rest at perception prices, which is the same trade §5.2 makes for the attack map at depth zero, applied at depth d .*

This is a transposition table, and saying so is not a concession. The point is where it lives. A searched fact is a pair of a configuration and the word of events that produced it, which is the object the history monad already supplies, and nothing in the design as previously specified used it. So the mechanism for shared search memory is the history monad, the transposition table is its ecological reading, and open problem Q4 — which was a loose end about the fifty-move and repetition rules — becomes the load-bearing question of how the society remembers what it has already imagined.

Observation 9.17 (What the society becomes, once it shares). Sixteen agents, each driving shadow boards to whatever depth it can fund, over an overlapping candidate space, against a common deposited record of what has already been searched, is a loosely-coupled parallel search sharing a transposition table. That is a familiar object; the engine literature calls the shared-memory version of it lazy SMP. What is unfamiliar here is the governance layer on top — which agent searches what is not assigned, it is advocated — and the fact that depth is per-agent rather than global.

Corollary 9.18 (The overhead is a constant, not an exponent). *Under Observation 9.17 the advocacy protocol costs a bounded number of rounds and a bounded number of messages per turn, independent of depth. It is therefore a constant factor on the turn, not a term that grows with β^d . Coordination overhead of that shape costs rating points; it does not make a player structurally weak, and §1.1 is written accordingly.*

Remark 9.19 (The cost question does not go away). Depositing a searched fact is not free and the deposit is not the world's gift. Whichever society is billed for it, the accounting must balance, and the interesting version of measurement M7 is now two-dimensional: who is billed for the attack map at depth zero, and who is billed for the search record at depth d . The framework's prediction in both cases is the same and the second is a much stronger test, because the duplication factor grows with β^d rather than staying at S .

10 Material as metabolism

The noughts arena routes and never mints: on a win the arena hands the winner the loser's metabolic name. In chess, capture is predation at ply granularity, and nothing needs reinterpretation for it to be so.

Definition 10.1 (Capture is harvest). When a ply's join terminates a piece-agent q , the capturing agent p receives q 's metabolic name; q 's remaining stack is thereby routed to p . No tokens are minted and none destroyed, so $\sigma + \kappa = \Theta$ holds across the ply.

Proposition 10.2 (Material advantage is budget advantage). *Under Definition 10.1, a side with more surviving material has, all else equal, both more advocates and more aggregate stack, hence can afford more perception per turn and more rounds of advocacy. The conventional evaluation term "material" becomes a statement about the society's computational capacity rather than a number added to a score.*

This is the most interesting thing in the design and also the least certain.

Conjecture 10.3 (Emergent piece values). *Let $\pi(t)$ be the average lifetime harvest — tokens acquired by capture, plus tokens the agent’s survival makes available to the society — of an agent of piece type t over a population of games. Then the ratios $\pi(\text{pawn}) : \pi(\text{knight}) : \pi(\text{bishop}) : \pi(\text{rook}) : \pi(\text{queen})$ approximate the classical $1 : 3 : 3 : 5 : 9$.*

We state it as a conjecture and not as a design goal, and the difference matters. Nothing in the architecture hands a piece its value. A queen is an agent whose move-generation contract reaches further, which means it can advocate for more candidates, bear on more squares, and belong to more coalitions; and under $\otimes = \min$ belonging to more coalitions is worth something only if the agent is available. If the classical ratios fall out, the framework has derived one of the few quantitative facts about chess that has no derivation. If they do not, the discrepancy is data about which of the framework’s pricing choices is wrong, and it is much more informative than agreement would be.

Remark 10.4 (Two more in the same currency). Offered as shapes rather than claims, because they are the sort of thing the design makes sayable and it would be a waste not to say them. A *pinned* piece is one whose budget is committed but unspendable, which is why Available zeroes it. *Zugzwang* is being obliged to spend a token when every available expenditure is loss-making, which in this setting is not a metaphor: the franchise is issued per round and a piece that must advocate for something is a piece that cannot decline the market.

10.1 The treasury

Harvest alone funds the society badly, and the defect is specific: it is procyclical. A side that is behind has fewer agents and smaller stacks, so by Principle 9.3 it calculates less far, so it falls further behind. A conventional engine has no such spiral, and an architecture that manufactures one should say whether it means to.

It does not have to. The noughts world already contains a second funding channel: the practice wall, a *source* rather than a prey — rate-limited, non-exclusive, non-adaptive and finite — and that access profile is what makes the world mixotrophic rather than purely predatory. All that is needed is to unfreeze its schedule.

Definition 10.5 (The treasury). The treasury is a source holding an opening reserve Θ_W , which is part of Θ and is never replenished. At each turn it disburses $q(\cdot)$ tokens per surviving agent of the moving colour, where q is non-decreasing over the game. Disbursement is non-adaptive: it does not read the position.

Conservation is untouched. Θ_W is fixed at the outset, the reserve drains monotonically, and $\sigma + \kappa = \Theta$ holds throughout, so the schedule is a redistribution of a fixed endowment over time and not a mint. A meagre opening allowance rising to a generous late one is exactly a large initial supply distributed increasingly as play proceeds.

Remark 10.6 (Key the schedule to material, not to ply). An exponential in ply number diverges against a finite reserve, so long games end in starvation and one harvests draws by exhaustion that are artefacts of the treasury rather than facts about the players. Keying q to *material captured* instead is bounded above, monotone, and self-normalising. It also says something better: the treasury pays out in proportion to how much of the position has been resolved, so a simpler position funds a deeper search, which is independently true of chess.

Proposition 10.7 (The endgame is where this society is richest). *Under Definition 10.5 with q keyed to captured material, the per-agent allowance is $q(c)/S$ with q non-decreasing and S non-increasing in captures. Per-agent budget therefore rises super-linearly as material comes off, and by Principle 9.3 so does affordable depth.*

Corollary 10.8 (A falsifiable signature). *Combining Proposition 10.7 with Proposition 7.10: in the endgame the society is simultaneously split-free by arithmetic and individually deepest by budget. The framework therefore predicts that this player is at its best in the endgame, which is the opposite of how weak engines usually fail, and the prediction is available from the first tournament.*

10.2 What the treasury must not pay for

The obvious refinement is to pay for good play: reward an agent whose move improved the position. It should be resisted, and the reason is not conservation, which a fixed reserve settles.

Principle 10.9 (No exogenous judge). *A disbursement rule that reads position quality is an evaluation function held by the treasury. Agents are then paid for agreeing with it, the ladder is selected for tracking it, and the yield measurement of §13 degenerates into a measurement of how well the ladder approximates the treasury's opinion. This is the leak Remark 5.5 guards against, arriving through the accounts rather than through the floor, where it is much harder to see.*

There is a version that pays for good play and keeps the judge endogenous, and the programme already specifies its ingredients.

Definition 10.10 (Assay-confirmed disbursement). An agent's draw on the treasury is conditioned on its own assay outcomes over the preceding turn: a confirmed hypothesis draws, a refuted one does not, and $\perp$ — the budget bottom — does not. The rule reads only the confirm/refute trichotomy the assay already emits, which is the same observable `Revise` uses to set its temperature.

Principle 10.11 (The question becomes the selection pressure). *Under Definition 10.10 a hypothesis that pays for itself buys its own depth, and one that does not starves its holder. "Is the yield of a hypothesis worth its price" stops being a statistic computed after the games and becomes the mechanism by which the population changes.*

This is the scientist note's motivation mechanism instantiated rather than approximated: token supply reduced when predictions degrade and replenished when they improve. It needs no judge, no exogenous score, and no new machinery, and it makes the experiment self-funding in the exact sense the note's central question asks about.

Remark 10.12 (Two components, two jobs). The schedule of Definition 10.5 sets the *floor* and removes the procyclical spiral, because it pays per surviving agent regardless of how that agent is playing. The rule of Definition 10.10 supplies the *differential* and is where selection lives. They should be built and measured separately: the first is a correction to a defect, the second is an experiment.

11 Where the policy lives

11.1 The extremes are not a dial

A piece's policy is the thing that maps the position to an advocacy. The question is where that map lives and whether it is priced. The sixteen-model-sessions extreme and the all-f1r3lang extreme are two answers, and they are not near-neighbours on a continuum.

A model session is opaque. Its choice function is not a clause, so it cannot sit in guard position and the assay construction does not apply. It is not priced, so cost accounting measures API calls rather than computation, and every measurement in §13 loses its denominator. It is not observable, so the coalition formulae become commentary on a decision

they did not make, and Remark 8.4 is false — a separate narrating faculty is exactly what a model wrapper reintroduces. At that extreme every structural result in this note evaporates and what remains is a multi-agent chess demonstration, of which there are many. Sixteen sessions per advocacy round is also slow enough that the round structure of §7.8 becomes unusable.

11.2 The aperture, which is the programme's own construction

None of that makes model-backed pieces incoherent, and the reason is that the choice-formulae note already worked out how strength enters a shard from outside. Its Route E is the *aperture*: a channel fed by the environment rather than by the term. Two things are said about apertures there. They are where choice of full strength can enter, and they are syntactically conspicuous, so a static analysis of aperture channels bounds a whole shard. Under the stratified *where* discipline, subject reduction holds and full-strength choice occurs exactly at declared apertures while the interior's choice type stays invariant.

A model session is an aperture, precisely and not by analogy. So a society with model-backed pieces is coherent in a stratified shard, and building it would *demonstrate* the stratification result on an artefact people care about rather than on a counterexample two lines long.

Principle 11.1 (Relocate, do not compromise). *The recommendation is not a midpoint between the extremes. It is that the model goes outside the ply loop by default, and inside only as a declared aperture in a labelled variant.*

Outside the ply loop there are two jobs a model is actually good at, and both are already roles in the existing design. The first is authoring and mutating piece policies between games, which is the *Revise* and *Crossover* and *Beget* role of the noughts listing, and which lands squarely on the reproduction note's crossover over quoted code — a model is a mutation operator on programs, which is a thing models do well. The second is reading the witness coalition out loud, which is post-hoc narration of a decision that was genuinely made elsewhere and is therefore honest.

Remark 11.2 (What the aperture variant would measure). Worth building second, not first, and worth building for a specific reason: it is the only configuration in which one can measure what a model's judgement is worth *in the framework's own currency*. Run the same society with j of sixteen pieces aperture-backed and $16 - j$ *f1r3lang*-backed, and the yield-per-unit metric prices the model's contribution against the ladder rungs. Nothing else in the current AI-evaluation landscape prices a model's judgement against a formally specified alternative on the same budget.

12 The implementation spec

This section is written to be handed to a developer. It fixes names, protocols, prices, interfaces, delivery phases and acceptance criteria. Where a decision is genuinely open it is marked OPEN rather than resolved silently.

12.1 Module inventory

Module	Society	Responsibility
Board	world	occupancy cells; the ply join; refusal of illegal joins
Settle	world	ray walks; deposit of <i>atk</i> , <i>pin</i> , <i>chk</i> , <i>ep</i> , <i>cast</i>
Arena	world	game lifecycle; harvest routing; adjudication
Clock	world	turn and round structure; resolver of last resort
Registry	world	the roll: which piece channels are live this turn
Franchise	player	issue of one linear advocacy token per live piece per round
Piece	player	per-agent policy: generate candidates, evaluate, advocate
Gate	player	the k -ary commit join, one instance per live candidate
Ladder	player	the formula catalogue; rung held per agent
Revise	player	per-agent rung revision from assay outcomes
Treasury	world	the source: schedule keyed to captured material; assay-conditioned draw
Metab	both	Live: reclaim, debit, reissue; death on insufficiency
Check	world	the coalition-logic and modal checker service (§12.6)

Check is placed in the world deliberately. It is an instrument the player *holds*, and its queries are charged to the caller, but it is not part of any agent's term; otherwise sixteen agents would carry sixteen copies of the checker and the caching requirement could not be met.

12.2 Namespace discipline

Every name below is manufactured by quotation, not *new-bound*, so every family is described by a predicate. *b* is the board channel, *pl* the player channel, *g* the game index.

Family	Name	Payload
board	@[*arena, g]	—
occupancy	@[*b, "occ", s]	0 or *p
attack	@[*b, "atk", s, c, p]	attacker descriptor
pin	@[*b, "pin", s]	pinning ray descriptor
check	@[*b, "chk", c]	checking piece set
en passant	@[*b, "ep", s]	capturable pawn channel
castling	@[*b, "cast", c, side]	right present or absent
king zone	@[*b, "kz", c, s]	s is in c's king zone
pawn structure	@[*b, "pwn", c, f, kind]	doubled / isolated / passed
piece	@[*b, "p", id]	—
piece stack	@[*b, "m", id]	σ , the metabolic stack
advocacy	@[*pl, "adv", @m]	advocating piece id
inhibition	@[*pl, "inh", @m]	inhibiting piece id
franchise	@[*pl, "V", id, r]	the round's linear token
commit	@[*pl, "commit"]	the chosen @m

Requirement 12.1 (No family is receivable by the wrong society). No contract of the player receives on `occ` except through the ply join of `Board`; no contract of the world receives on `adv`, `inh` or `V`. This is Proposition 5.2 discharged as a code review checklist item, and it is the single most important invariant in the system.

12.3 The turn protocol

A turn proceeds as follows. Steps marked $\star$ are metered to the player; steps unmarked are metered to the world.

1. **Settle.** `Settle` walks every ray from every occupied square and deposits the `atk` family; then walks outward from each king and deposits `pin` and `chk`. Terminating and confluent by Remark 5.3. On completion it signals the clock.
2. **Roll.** `Registry` publishes the set of live piece channels for the moving colour.
3. **Disburse.** `Treasury` draws $q(c)$ from its reserve for each live agent, with c the material captured so far and q non-decreasing (Definition 10.5); under assay-conditioned disbursement (Definition 10.10) an agent whose last assay refuted or returned $\perp$ draws nothing. The treasury never reads the position.
4. **Franchise.** `Franchise` issues one linear `V` token per live piece into that piece's located stack. Round counter $r := 1$.
- $\star$ **Generate.** Each `Piece` enumerates its own pseudo-legal moves from its own square by its own move contract. No piece enumerates another's. The union over pieces is the candidate set; nobody computes the union.
- $\star$ **Evaluate.** Each `Piece` evaluates its candidates against the rung it currently holds, by querying `Check`. The query is an assay: the formula sits in guard position, so evaluating is testing.
- $\star$ **Advocate.** Each `Piece` spends its `V` token on $adv(m)$ for its best candidate m , or declines. OPEN: whether a piece may advocate for a move it does not itself make — see §12.9.

5. **Gate.** A `Gate` instance stands on each live candidate. The first to accumulate k advocacy sends fires and writes `@m` to `commit`. By Theorem 7.6 at most one fires when $k \geq \lceil S/2 \rceil$.
6. **Round or resolve.** If no gate fired and $r < R$, then $r := r + 1$ and control returns to step 4, with unconsumed advocacy left standing and readable. If $r = R$, the clock applies Definition 7.9.
7. **Ply.** `Board` performs the N -ary join of Definition 6.1, guarded on the constraint family. A refused guard leaves the message in the tuple space, which is the platform's own semantics for a failed `where`, and the clock retries with the next-best candidate.
8. **Harvest.** If the join terminated an agent, its metabolic name is routed per Definition 10.1.
- ★ **Revise.** Each piece updates its rung from its own assay outcomes by `Revise`. Refutation rate is the temperature, as in the noughts world; it is an observable the assays already emit rather than a schedule imposed from outside.
9. **Adjudicate.** `Arena` tests for mate, stalemate, and the draw conditions, and either ends the game or passes the turn.

Remark 12.2 (Where the cost actually goes). Steps 5 and 11 are the player's whole expenditure and step 1 is the world's. That is the entire content of measurement M7. Note that step 4 is charged to the player but is cheap and irreducible — a piece must know its own moves — and step 7 is charged to the world because a join firing is a COMM, which under the native integration is metered at the reducer on the deploy envelope.

12.4 The ladder

Each piece holds a rung. Rungs are cumulative and evaluated highest-priority first, as in the noughts listing. `Emp`, `Bears`, `Valuable`, `Available`, `Central` and `Develops` are structural predicates over the deposited families; $\langle\langle K_m \rangle\rangle$ is the settled modality at the ply m .

Rung	Character	Formula added	κ
0	—	$\top$ (any legal move)	0
1	non-modal	$\exists m. \text{Capture}(m) \wedge \neg \text{Defended}(to(m))$	2
2	non-modal	$\vee \neg \text{Hangs}(m)$: the destination is not attacked by a cheaper piece	2
3	spatial	$\vee \langle\langle K_m \rangle\rangle \text{Fork}(me)$, Definition 8.7	4
4	alternating	$\vee \forall m'. \langle\langle K_m \rangle\rangle [K_{m'}] \neg \text{Fork}(them)$	8
5	spatial, depth 0	$\vee \text{Central}(to(m)) \vee \text{Develops}(m)$	1
5a	spatial, depth 1	$\vee \langle\langle K_m \rangle\rangle \text{KingSafe}(me)$, Definition 12.3	2
5b	spatial, depth 1	$\vee \neg \text{Worsens}(m, \text{Pawns})$, Definition 12.4	2
6	forcing	$\vee \exists s. \langle\langle K_m \rangle\rangle \text{Forces}(s)$, Definition 9.10	2^d
χ	fixed point	$\nu X. \forall m. [K_m](\neg \text{Lost} \wedge \exists m'. \langle\langle K_{m'} \rangle\rangle X)$	32

The prices are the noughts schedule $\kappa(d) = 2^d$ per candidate examined, charged by the depth of the formula the rung installs. For the modal rungs this is no longer a stipulation:

by Proposition 9.8 it is the cost of instantiating a shadow board and driving it d plies, with $\beta = 2$ continuations per ply. Rung 6 is priced 2^d rather than at a fixed constant because it is a ν over the run tree and its depth is whatever the caller funds, which is Principle 9.3: the rung an agent holds fixes the *shape* of what it asks, and its stack fixes how far. Rung 5 is priced at 1 because it is name-level depth zero, and that pricing is what made it the best marginal purchase in the noughts measurement; whether it repeats here is M2.

Rungs 5a and 5b are new in this draft and they are the reason §1.1 moved leaf coarseness onto the chosen list.

Definition 12.3 (King safety). Let the settling deposit a king-zone family $kz(c, s)$ marking the squares adjacent to and in front of c 's king — one additional walk, over a neighbourhood of bounded size, in the pass that already walks every ray. Then

$$\text{KingSafe}(c) := \langle E \rangle_{N_{atk}}^V \lambda p. (\exists s. kz(c, s) \wedge \text{Bears}(p, s) \wedge \text{Colour}(p, \bar{c}))$$

evaluated as a graded count over the attack namespace restricted to the zone. It is a read over deposited markers at spatial depth one.

Definition 12.4 (Pawn structure). Let the settling deposit $pwn(c, f, kind)$, with *kind* ranging over doubled, isolated and passed, computed from occupancy over the file namespace

$$@[*b, "occ", (f, \cdot)]$$

and over its adjacent files. Doubled is a multiplicity condition on one file, isolated a negation over the two adjacent files, passed a negation over three. $\text{Worsens}(m, \text{Pawns})$ compares the deposit before and after m and is therefore a depth-one read under $\langle\langle K_m \rangle\rangle$.

Remark 12.5 (Expressibility was never the constraint; price is). A previous draft treated the coarse ladder as forced, on the grounds that a rung must be a formula of the generated logic. That confuses a short ladder with a weak logic. The logic carries ν , graded separation over described namespaces, the name predicate, and redex-indexed modalities, and under reflection its reach is wider again — the programme's own history here is one of *restricting* it deliberately and then relaxing the restriction, which is not the profile of a structural bound. What differs between predicates in this setting is not whether they can be said but what they cost, and cost turns on what the world has deposited. Definitions 12.3 and 12.4 are read-versus-drive (§5.2) applied to evaluation rather than to legality: classically these are computed per evaluated position, and deposited they are read once per agent at depth one. What remains genuinely open is neither expressibility nor price but *findability*, which is Q11.

Note also, per Proposition 9.11, that rung 4 is Forces at fixed-point depth one with a weaker target predicate, so the top of the ladder is a refinement chain rather than a pair of competing instruments.

Remark 12.6 (χ is unreachable, and that is the point). In noughts χ was exact at nine unfoldings because the board loses a square every step. Chess has no such bound. χ is therefore listed for completeness and as a limit object; no agent will hold it, and by Remark 3.1 that is exactly the condition under which the ecology stays alive.

12.5 The cost schedule

Operation	Grade	Price
read one deposited marker	perception	1
evaluate a rung- d formula, one candidate	reflection	$\kappa(d) = \beta^d, \beta = 2$
one shadow ply (instantiate and drive)	reflection	1
cached query, unchanged deposit	perception	1
enumerate own pseudo-legal moves	perception	1 per square walked
spend an advocacy token	interaction	2
fire a commit gate	interaction	$2k$
perform a ply join of arity n	interaction	$4n$
one ray step in <code>Settle</code>	world	1
harvest	interaction	0 (routing, not work)

The non-modal prices here are CHOICES, not derivations, and the note should say so in the same breath as reporting any number computed with them. The noughts work varied its price scale over three settings and found the shape of its results robust and the boundaries not; that sensitivity analysis is required here for the non-modal rows and is part of the acceptance criteria in §12.8. The modal rows are exempt by Proposition 9.8: their price is architecture, and the one free parameter is β , which is measured rather than set — it is what pruning (§9.4) reduces, so reporting β per configuration is itself a result.

12.6 The checker interface

The service assumed in §2.4 is used through one contract.

```
// verdict in {"t", "b", "f"}; witness is Nil, a redex position, or a
// coalition (a list of piece ids); cost is charged to the caller's stack.
contract Check(@formula, @posHandle, @algebra, @depth, payer, ret) = {
  // ret!((verdict, value, witness, cost))
  // witness at depth d is a PRINCIPAL VARIATION (Prop. 9.13), and at
  // depth 0 a coalition (Cor. 8.3). Shadow boards are manufactured from
  // posHandle by quotation and driven privately (Def. 9.5).
}
```

formula

a quoted formula of the generated logic extended with $\langle E \rangle_N^V$, with namespaces given as name predicates.

posHandle

a handle on the current deposit, not a copy of the position. The checker peeks at the live families and never consumes them; when a modal rung requires plies to fire, it manufactures a shadow board from this handle by quotation and drives that (Definition 9.5). This is Requirement 2.1 enforced by the interface shape.

depth

the ply budget for the query, which by Principle 9.3 is set by what the caller can pay rather than globally. A query whose funds run out mid-search returns "b" with the best witness found, which is the budget bottom in its search-shaped form.

algebra

one of "B", "Rnn", "trop", "vit". Witnesses are returned for the last two.

payer

the caller's metabolic channel. A caller that cannot fund the query receives verdict "b", which is the budget bottom and is deliberately indistinguishable from a stuck environment.

Requirement 12.7 (The checker may not move *the* board). On the live board b the checker must peek and never consume: it may receive on no family of b , including `occ`. Violating this makes the pride move the zebra by looking at it, in the phrase the herd note used for the same hazard, and it is the reason the noughts listing insisted on `peek` rather than a consuming receipt in its own scan.

It may receive freely on the families of any shadow board it has manufactured, and for the modal rungs it must, since $\langle\langle K_m \rangle\rangle$ requires the ply to fire. See Requirement 9.6, which is the operative form; the separation is structural, because a shadow name is a quote of a term properly containing the checker and is therefore fresh in b .

12.7 Reproduction and the offline loop

Between games the population evolves. This is the noughts world's `Pick / Crossover / Beget` apparatus and it transfers with one change of emphasis. There are two loci, the rung and the policy, and the noughts measurement showed that an individual can reach the first by deepening and cannot reach the second at all — the opening changed only across generations, by crossover, because it is a move of distance 2^{-1} in the ultrametric and an individual's incremental revisions are confined to their initial ball.

The chess analogue of “the opening” is the piece's *advocacy style*: how readily it advocates, what it treats as valuable, whether it defers to standing advocacy. That is the locus crossover must reach, and it is where a model-backed mutation operator earns its place (§11).

12.8 Delivery phases

P0 — the world alone.

Board, Settle, Registry, Clock, Arena. Acceptance: legal move generation validated against a standard perft suite to depth 5 from the initial position and from a set of tactical positions; the settling shown terminating and confluent by test; every ply a single join.

P1 — one agent, no society.

A single `Piece` contract driving all pieces in turn, rung 0–2, no advocacy. Acceptance: plays legal complete games against a fixed reference opponent; metering totals reconcile against $\sigma + \kappa = \Theta$.

P2 — the society, arithmetic guards.

Sixteen agents, franchise, commit gates, rounds, last resort. Rungs 0–2 and 5 only, since those are the non-modal rungs and are expressible in the live guard fragment. Acceptance: Theorem 7.6 confirmed empirically at k below and above $S/2$; last-resort firing rate reported.

P3 — the checker, and lookahead.

`Check` integrated with shadow boards (Definition 9.5); rungs 3, 4 and the forcing rung 6 enabled; per-agent depth funded from the agent's own stack. Acceptance: Requirement 9.6 verified by showing that no contract receives on a live-board family; β measured and Proposition 9.8 checked against the metered cost of a rung- d query; M1, M2 and M9 produced.

P3a — shared search memory.

The searched-fact deposit of §9.5, over the history monad. Split out from P3 because it is optional for correctness and decisive for latency. Acceptance: the depth- d half of M7.

P4 — the economy.

Piece stacks, harvest, metabolic death, the `Treasury` with its material-keyed schedule, the offline reproduction loop. Acceptance: M6 and M11 produced; conservation audited per ply, including the treasury draw, against a fixed $\Theta = \Theta_W + \sum_p \sigma_p(0)$.

P4a — selection.

Assay-conditioned disbursement (Definition 10.10), built and measured separately from the schedule per Remark 10.12. Acceptance: the population's rung distribution reported over generations, since this is where Principle 10.11 either bites or does not.

P5 — the aperture variant.

j of sixteen pieces backed by a model session through a declared aperture, in a stratified shard. Acceptance: M8 produced; the static analysis of aperture channels demonstrated.

P0 through P2 are buildable today on `feature/cost-accounted-rho`. P3 depends on the assumed checker. A syntactically manipulable `Theory` of positions is *not* on the critical path and should not be put on it: the DDL and module syntax are absent from the branch, which is the same dependency F1R3Skein is waiting on, and nothing in P0–P5 requires them.

12.9 Open specification items

S1

May a piece advocate for a move it does not itself make? Yes gives genuine coalitions in the advocacy structure and lets a defender back an attacker's plan; no keeps the franchise interpretation clean and makes advocacy self-interested. Our inclination is yes, because it is what makes the advocacy set a coalition rather than a poll, but it changes Theorem 7.6 not at all and can be decided late.

S2

Is k absolute or a fraction of S ? Proposition 7.10 says this determines whether the endgame is split-free or gate-dead.

S3

Cross-inhibition in or out of the first build? Recommended out, per §7.

S4

Is the opponent a society too, or a conventional engine? Recommended conventional for P1–P4: a fixed yardstick is needed, and the asymmetry is itself the experiment. Society-versus-society belongs after M1.

S5

Do pieces hold individual stacks from P2, or only from P4? Holding them early makes the metering real but couples two sources of variance.

S7

The shape of q . Linear in captured material is the conservative default; the interesting question is how convex it may be before the opening becomes unplayably shallow, and whether Θ_W should be sized so that the reserve is nearly exhausted at a typical game's end or deliberately not.

S8

Does an agent's treasury draw go to its own stack or to a common pool the society spends collectively? The first keeps the metabolism individual, which is what makes harvest meaningful; the second would make depth a society-level resource and is the closest this design could come to a global depth parameter, which is a reason to be suspicious of it.

S6

What is the reference opponent? It must be weak enough that the society is not simply crushed and fixed enough that measurements are comparable across phases.

13 The measurements

Eight measurements. Each states what would be reported and, where it matters, what result would embarrass the framework.

M1 — the yield ladder, against depth.

For each rung, win/draw/loss against a fixed reference opponent at fixed strength, and cumulative metered cost, over a fixed number of games. Reported as the noughts table was: win rate, cost, and marginal return per metered unit — but indexed by *affordable depth* as well as by rung, because by Observation 9.1 the rung fixes the shape of the question and the stack fixes how far it reaches, so a rung index alone does not identify what was purchased. The primary artefact is the curve of yield per metered unit against depth. Unlike noughts these are sampled estimates with confidence intervals, not exact counts, and the note reporting them must say so in its own results section.

M2 — non-modal share, deconfounded.

The fraction of total yield delivered by rungs 1, 2 and 5 and the fraction of total cost they consume, against the noughts figures of 92.3% and 34.3%. This is the discriminating measurement of Principle 8.6, and as stated in an earlier draft it was confounded: the noughts figure compares against modal rungs that are two-ply search on a 5478-position game, and the chess figure would compare against modal rungs that are d -ply search on chess, so the two ratios are not ratios of the same thing. The comparison must be made at matched depth, and the result reported as a family of ratios indexed by d rather than as one number. If the non-modal share falls monotonically in d , read-versus-drive is a statement about shallow play; if it holds up, it is a statement about games.

M3 — the coalition rung.

Rung 6's marginal return, and separately whether its witness is *informative*: does the returned coalition coincide with what a strong player would name? Scored by blind comparison on a set of tactical positions. The herd note found that the modality bought irreversibility rather than accuracy — precision 1.0000 at every sampled time, at low coverage — and if that repeats, the yield metric underprices instruments that decline to answer and the second-currency principle acquires its second instance.

M4 — split and deadlock against k .

Firing rate of the commit gate, split rate, deadlock rate and last-resort rate, over k from 2 to 16 at $S = 16$. Theorem 7.6 predicts split rate identically zero for $k \geq 9$, and the prediction is structural, so a single observed split above that threshold is a bug in the franchise's linearity and should be treated as such.

M5 — the endgame crossing.

Decision character as S falls through $2k$ within single games: rounds to commit, advocacy entropy, and last-resort rate, bucketed by material remaining. This is the measurement Remark 7.11 exists for and the one we would report first, because it is the only prediction here that the source domain could not have made.

M6 — emergent values.

Lifetime harvest per piece type against the classical 1 : 3 : 3 : 5 : 9, per Conjecture 10.3. Report the raw ratios and the regression, and report disagreement loudly if it occurs.

M7 — who is billed, at two depths.

Total metered cost per game under the world-billed and player-billed attack-map architectures of §5.2, split by society; and, separately, with and without the shared search memory of §9.5. The framework makes the same prediction twice, and the second is much the stronger test, because the duplication factor at depth zero is bounded by S while at depth d it grows with β^d . This is the cheapest experiment in the set and it directly tests the read-versus-drive thesis.

M8 — the aperture's price.

With j of sixteen pieces model-backed, yield per metered unit as a function of j , against the all-f1r3lang baseline at matched depth. Prices a model's judgement in the framework's currency, and at matched depth it prices it against a stated amount of search, which is the comparison the current AI-evaluation landscape does not make.

M9 — what pruning buys.

The measured β per configuration, with and without a pruning discipline, and the resulting shift in the M1 curve. By Corollary 9.14 a sound discipline leaves the attaining witness alone, so any change in the principal variation under pruning is a soundness bug and should be treated as one rather than as noise.

M10 — aggregation, netted against coverage.

Two quantities, reported together, because separately either one misleads. The *commit gap* is the per-turn difference between the value of the move the society played and the best move any single agent advocated. The *coverage gain* is the difference between the value of the best move any agent advocated and the best move the strongest single agent would have found running alone at its own depth with its own candidate cap. The first is what the governance layer costs; the second is what sixteen heterogeneous searchers buy. The net is the only honest answer to whether plurality beats maximum here, it is the number a sceptical reader will ask for first, and §1.1 declines to predict its sign.

M11 — the endgame signature.

Playing strength and affordable depth as functions of material remaining, against the reference opponent, with and without the treasury of §10.1. Corollary 10.8 predicts that this player is relatively *strongest* in the endgame, which is the opposite of the usual failure profile of a weak engine and is therefore easy to falsify. Reported alongside M5, which measures the decision-procedure half of the same crossing.

Remark 13.1 (The counter-move, if there is time). The herd note's most striking result was the counter-move: an adversary that introduces seams into the population does not induce *error* in the observer, it denies *liveness* — coverage collapsed from 0.7520 to 0.0115 while precision stayed above 0.9986. The chess analogue is an opponent that plays to reduce the society's advocacy coherence rather than to win material, and the prediction is that such

an opponent should drive the society to last resort rather than to blunders. If it holds, illegibility as denial of liveness acquires an adversarial instance, and the legibility trade — that a population buys privacy with the same currency it buys agreement — becomes something one can demonstrate rather than argue.

14 What will go wrong

The one candidate for a forced limit, and it is not established. A conventional engine takes the maximum over its candidates. A quorum takes a plurality above a threshold, so there is a gap at the commit: a piece holding the right move loses to eight pieces holding a worse one. This is architecture rather than defect — the whole point of §7 is that a maximum over sixteen agents requires an evaluator none of them can afford to be — and earlier drafts of this note therefore filed it as a structural weakness.

That filing was premature, because the same architecture supplies a countervailing gain and no draft has netted them. Sixteen agents at heterogeneous depths (Principle 9.3) over different candidate sets examine a wider tree than any one of them could, which is Observation 9.17 read as a benefit rather than as an overhead. Whether the commit gap exceeds the coverage gain is not known, it is not obvious in either direction, and M10 is specified to measure exactly the difference. Until it does, the honest statement is that this note knows of no established structural bound on the society’s strength.

What is chosen, and what each costs. Four dials, none structural, each set conservatively in this spec and each worth varying.

- **The ladder’s length.** Rungs 5a and 5b were added to this draft in an afternoon and the ladder is still short: no mobility term, no notion of the initiative, nothing about rook activity or the bishop pair. Deep search over a coarse evaluation is weak search, and the remedy is more rungs and more deposited structure, not more logical power (Remark 12.5). What is not known is whether the revision process can *find* good rungs, which is Q11 and is a fact about the learner rather than about the language.
- **The treasury schedule.** §10.1 removes the procyclical spiral that harvest-only funding creates; without it the side that is behind calculates less far and falls further behind. That spiral was reported as structural in an earlier draft and is not.
- **The round budget R and the candidate cap per agent.** Both *change the science*: capping candidates is capping $\exists$, and capping depth is capping the fixed point. Report them with every number.

The honest expectation. The evaluation criterion must accordingly be yield per metered unit at a *stated* affordable depth, with playing strength as a secondary sanity check, fixed before the first game rather than discovered in the results. A project of this kind fails publicly in exactly one way, which is by having promised strength.

The society is small. The swarm results are asymptotic in scout population; at $S = 16$ the regimes are coarse and the advocacy distribution is lumpy. This cuts both ways, since it is also what makes Proposition 7.10 reachable, but no fine-grained result about quorum dynamics should be expected from sixteen agents.

The guard fragment. Structural and modal guards are not executed by the live reducer; they are specifications. P2 is buildable because rungs 1, 2 and 5 are expressible arithmetically over deposited markers, which is a deliberate design of the ladder rather than a happy accident, but P3 is gated on the assumed checker and the note should not pretend otherwise.

Latency. Sixteen agents, up to R rounds, a checker query per candidate per agent, and each modal query driving β^d shadow plies. Even in the all-f1r3lang configuration a turn is many thousands of COMMs, and at rung 6 many more. A demonstration has to be watchable, and the three dials above are what make it so, which is the practical reason they need tuning early. Shared search memory (§9.5) is the one mechanism that buys latency back without turning any of them down, which is an argument for bringing it forward rather than leaving it to Q4.

Reflection, a fifth time. The subject-reduction hazard the programme has now met four times is present here too: a received process can carry a formula of higher altitude than the receiving clause declares, and this society passes quoted moves and quoted formulae between agents constantly. The stratified *where* discipline is the known repair and it should be adopted from P2 rather than retrofitted, because retrofitting it means revisiting every advocacy message.

15 What is shown and what is not

Shown, in the sense of proved here from imported premises: Proposition 5.2 (perception must not act), Proposition 6.2 with Corollary 6.3 (the join taxonomy and the survival of $\langle K_j \rangle$), Proposition 8.1 and the consequent choice of quorum algebra, Theorem 7.6 with its corollary that $k \geq \lceil S/2 \rceil$ buys structural no-split, Proposition 9.8 (the modal price scale is derived from shadow driving, not chosen), Proposition 9.11 (rung 4 is the forcing rung at depth one), and Proposition 9.13 with Corollary 9.14 (the witness at depth d is a principal variation, and pruning is sound exactly when attainment survives it). Theorem 7.6 is the only genuinely new theorem in the note and its proof is three lines, which is the correct length: the work was done by choosing a calculus in which linear tokens and N -ary joins are primitive.

Argued but not proved: that the world-billed architecture wins (§5.2, awaiting M7); that legality belongs to the world (Principle 5.4, which is a design principle with a cost argument behind it, not a theorem); that the model belongs outside the ply loop (§11); and Principle 9.3, that endogenous per-agent depth is a feature rather than a handicap, which §14 now classes as a *chosen* dial rather than a structural property and which M1 is built to expose against the per-turn-allowance variant.

Conjectured: emergent piece values (Conjecture 10.3).

Not addressed at all, and the first of these is the important one. *A good evaluation function.* The ladder is an evaluation function and it is still a thin one, even with rungs 5a and 5b. What this note now contains is a *method* for extending it — deposit the structure, read it at depth one — and no demonstration that the method reaches modern quality, nor any account of how the society would discover the extensions for itself, which is Q11. Everything claimed about yield is therefore claimed for a thin evaluator, which is the right scope for the measurement and the wrong scope for any claim about strength. Also not addressed: opening theory and endgame tablebases, both external knowledge the society has no principled way to consume; the fifty-move and repetition rules, which are history-dependent

and want the history monad rather than the deposit; and time control, which would make the budget a clock and is probably the most interesting thing omitted.

16 Open problems

- Q1 Does the noughts yield result survive? This is M2 and it is the project's reason for existing.
- Q2 Is there a principled k ? The design fixes $k \geq \lceil S/2 \rceil$ for safety, but nothing says which value in that range is best, and the swarm's answer — that k is not a number but a rate threshold, scaling with encounter geometry — suggests the chess analogue is a rate as well.
- Q3 Does agent-per-plan beat agent-per-piece? Remark 4.2. The obstacle is that a plan cannot be captured, so predation would have to be redefined.
- Q4 How does the society remember what it has already imagined? A searched fact (Definition 9.15) is a configuration paired with the word of events that produced it, which is the history monad's own object, and the transposition table is its ecological reading. The fifty-move and repetition rules fall out of the same construction, which is how this question came to be asked. It is now the largest open item in the design rather than a loose end, because by Remark 9.19 the duplication it addresses grows with β^d .
- Q5 Does the coalition modality's irreversibility-not-accuracy signature repeat? M3.
- Q6 Can the society learn its own `Open`? The herd note left this open in general; in chess the analogue is whether the pieces can discover which subsets of themselves are worth treating as coalitions, rather than having the namespace predicate handed to them.
- Q7 Does subject reduction hold for the stratified discipline in the presence of quoted moves passed as advocacy payloads? The fifth appearance of the same suspect.
- Q8 Is there a composition law for societies? Two coalitions that each bear on a square are not obviously composable, and the composing-learners note's disjointness-versus-overlap dial is the obvious place to look.
- Q9 What does time control do? Making the budget a clock rather than a stack changes the metabolism from conservation to flow, and it is not obvious the conservation results survive.
- Q10 Does the society's play have a signature a human can recognise? Not a formal question, but the one a reader will ask, and the witness coalitions of Corollary 8.3 are the only handle on it.
- Q11 Can the revision process *find* a good evaluation? Not whether the logic can express one — it can, and §12.4 exhibits two more terms — nor whether it can afford one, since deposit makes these depth-one reads. The open question is findability, and the programme already has a result pointing the wrong way: confinement says an ultrametric walk of steps all below 2^{-n} never leaves that ball, so a purely incremental scientist cannot escape its initial basin, and the noughts measurement showed this concretely when the best opening reversed under deepening and the better variant proved reachable only by crossover across generations. Choosing the *shape* of an evaluation is a distance- 2^{-1} move, so no individual gets there. If that holds, the population and

its crossover operator are the only route, which is precisely where the model-backed mutation operator of §11 earns its place. This is the most consequential question in the note, it is not about chess, and it relocates the programme’s real limit from the hypothesis *language* to the hypothesis *search*.

Q12 Once M10 has priced the aggregation loss, is there a commit rule that reduces it without reintroducing a chairman? Weighted advocacy, a second round restricted to the top candidates, and a veto token are all cheap to try; each buys back some of the maximum and each gives something up, and what they give up is exactly what Theorem 7.6 was protecting.

17 Conclusion

The proposal was to make each piece an agent. The analysis above says that this is a well-formed thing to do in cost-accounted ρ , that three of its apparently difficult parts are already solved by constructions in hand, and that one of them is easier here than in the setting it came from.

The board encoding wants three name families rather than one, because perception must not be able to act and because the attack map is the shared expensive object whose billing is the project’s cheapest experiment. A ply is an N -ary join and the join arity is the move taxonomy, which is what keeps the generated modality generated. Coordination is quorum sensing rather than consensus, and the argument for that is a derivation from budget rather than a design preference. Legality is the one unavoidable universal and it belongs to the world, which has the pleasant consequence that pin-awareness becomes perception.

There is one thing an earlier draft of this note had backwards, and it is worth recording as a correction rather than quietly fixing, because the mistake is instructive. That draft treated the context-labelled modalities as expensive evaluation predicates and then conceded that the society could not search. But modal depth is ply depth, χ is game-tree search written as a greatest fixed point, and $\kappa(d) = 2^d$ was branching cost under another name. The society searches; what is unusual is that its depth is endogenous, differs between agents, and is funded by material. Lookahead is reflection — quote the world and drive the copy — which gives the calculus’s oldest primitive a second job in this programme and gives the scientist note’s imagination a mechanism it lacked.

The part that is easier here than in the swarm is the two-quorum theorem. In the bees it is a structural impossibility result confirmed by simulation. In ρ it is arithmetic: a quorum is a join arity, one linear franchise token per piece per round is the polity’s dereliction, and two k -ary joins over a pool of S linear tokens need $2k \leq S$. A designer who sets $k \geq \lceil S/2 \rceil$ has bought a safety property outright, for any policies whatsoever, and has thereby confined cross-inhibition to the single job of breaking deadlock, where a cheaper mechanism — reading the standing advocacy between rounds — is available.

What the project is for is narrower than what it will look like. It will look like a chess engine, and this note has stopped predicting how good one. Three successive drafts located a structural limit and each was wrong: not that the society cannot search, since the modalities do the lookahead; not that its funding is procyclical, since the treasury is a schedule and schedules are set; not that the logic cannot express a real evaluation, since it can and price rather than expressibility is what governs. What is left is one candidate, the commit gap, and it is offset by a coverage gain nobody has netted. The honest position is that the note knows of no established bound, and that this is a weaker and more useful statement than either of the confident ones it replaces.

It is worth adding what should have been said at the start. Yield and strength are different quantities — the noughts result *is* that they come apart, with the truth itself dominated on cost — so nothing about this player’s rating answers the question the project exists to ask. It is an instrument for that question: whether the finding that non-modal rungs deliver most of the yield for a third of the cost is a fact about games or a fact about a game small enough to solve — and, now that the modal rungs are correctly identified as search, that question is properly asked at matched depth, as a family of ratios rather than a single number. Noughts-and-crosses cannot answer that, because everything about it is exact and nothing about it is hard. Chess can, and the same board that answers it also happens to be a world where the ecology cannot die of its own perfection.

A The listing

Conventions follow the game note. A parameter written `c` is a NAME; a parameter written `@d` is DATA; a channel is passed as `*c`, since `@(*c) = c`. Guards marked `///!` ideal use structural or modal connectives that the current reducer does not evaluate; they are specifications of a check rather than checks. Everything else runs. The listing is the spine only: move generation per piece type, the full settling walk and adjudication are elided, since they are long and unsurprising, and the file `chess.rho` accompanying this note carries them.

```
// =====
// A SOCIETY OF PIECE-AGENTS THAT PLAYS CHESS
// =====
//
// Companion listing to "Sixteen Scientists on One Board". Every construction
// of the note appears here as ordinary code:
//
// Sec. 5 the board -- Board, Cells, Settle, Ray (three families)
// Sec. 5.3 legality -- Legal, Pins (the world's ALL)
// Sec. 6 the ply is a join-- Ply (N-ary, atomic)
// Sec. 7 coordination -- Franchise, Advocate, Gate (quorum = arity)
// Sec. 7.6 rounds -- Clock (+ last resort)
// Sec. 8 coalition rung -- Ladder rung 6 (///! ideal)
// Sec. 9 lookahead -- Check, Shadow (reflection)
// Sec. 10 material -- Harvest, Live (conservation)
// Sec. 10.1 the treasury -- Treasury, Disburse (a source)
//
// CONVENTION. A parameter written 'c' is a NAME (a channel); a parameter
// written '@d' is DATA. To pass a channel one sends '*c', since @(*c) = c.
//
// NOTATION. Guards are written in the generated spatial-behavioural logic
// extended with the coalition modality: out(n)phi, in(n)phi, phi | psi, the
// name predicate @phi, the context-labelled modality <K_m>phi, its settled
// form <<K_m>>phi, the greatest fixed point nu X. phi, graded separation
// phi |_k^N over a DESCRIBED namespace N, and <E>_N^V psi over opens of N.
// There is no hidden-name quantifier and no need of one: 'new' is syntactic
// sugar, since a name is a quoted process and no process contains the name
// that codes it, so a name fresh in P is manufactured by quoting any term
// properly containing P. Every family below is manufactured that way, which
// is what makes it DESCRIBABLE and hence what makes |_k^N and <E>_N^V apply.
// The current reducer accepts only the arithmetic and pattern fragment of
// 'where'; guards using a structural, modal or coalition connective are
// marked '///! ideal' and specify a check rather than perform one.
//
// LOOKAHEAD IS REFLECTION (Sec. 9). A naive reading of this architecture
// says search is beyond it: sixteen budget-bounded agents, a rendezvous
```

```

// price per candidate, no central evaluator, in a game that rewards depth.
// That reading ignores what the context-labelled modalities are for. The
// modalities are not an evaluation device standing beside search; they ARE
// the search, and modal depth is ply depth. <<K_m>>phi requires the ply to
// FIRE, so evaluating a modal formula means exhibiting the rewrite -- which
// the checker does on a SHADOW BOARD @[*chk,H,d,i], manufactured by quoting
// its own channel together with a handle on the deposit. It may drive that
// without restriction because it is its own term, and by freshness-by-
// quotation
// quotation it cannot collide with the live board. Imagining a position is
// quoting the world and driving the copy.
//
// With the shared search memory of Sec. 9.5 the society is recognisably a
// loosely-coupled parallel search over a common record (Obs. 9.17), so the
// advocacy protocol is a CONSTANT FACTOR per turn and not an exponent
// (Cor. 9.18). What makes this player weak is elsewhere: the leaf evaluation
// below, and the fact that a quorum aggregates by PLURALITY where an engine
// aggregates by MAXIMUM.
//
// THE THREE FAMILIES (Sec. 5.1). Square s of board b carries:
// occ(s) = @[*b,"occ",s] occupancy; the ONLY family a ply consumes
// atk(s,c,p) = @[*b,"atk",s,c,p] deposited attack markers; read-only
// pin(s) / chk(c) / ep(s) / cast(c,side) deposited constraints
// Were attack markers deposited on occ, a scan signalling "p bears on s"
// could rendezvous with a move protocol receiving on occ(s) and the scan
// would PLAY A PLY. Perception must not be able to act; the separation of
// access grades one and two is enforced here by a choice of namespace, and
// with distinct families the failure is UNREPRESENTABLE rather than merely
// avoided by discipline.
//
// CONSERVATION. Nothing mints. The Treasury holds an opening reserve Theta_W
// that is part of Theta and is never replenished; its schedule redistributes
// a fixed endowment over time. Franchise issues one LINEAR advocacy token per
// live piece per round and does not carry it over. Harvest MOVES a captured
// agent's stack into the capturer's. Every Live debit moves a token from
// sigma into kappa. So sigma + kappa = Theta throughout.
// =====

new stdout('rho:io:stdout'), display, arena, wall, player, clock,
  Live, Treasury, Disburse, Board, Cells, Ray, Settle, Pins, Legal, Ply,
  Harvest,
  Registry, Franchise, Piece, Candidates, Evaluate, Advocate, Gate,
  Ladder, Revise, Check, Turn, Round, Adjudicate, Boot
in {

  for (@line <= display) { stdout!(line) }

  // =====
  // METABOLISM -- reclaim, debit, reissue.
  // Between the receipt and the send the stack is a message in flight: living
  // is racing others to your own stack, and the metabolic channel m is
  // exactly the vulnerability that Harvest exploits when a piece is captured.
  // =====

  | contract Live(m, @burn, step) = {
    for (@sigma <- m) {
      match sigma > burn {
        true => { m!(sigma - burn) | *step | Live!(*m, burn, *step) }
        false => { display!("DIED", "could not fund a step") }
      }
    }
  }
}

```

```

// =====
// THE TREASURY -- a source, not a prey. Sec. 10.1, Def. 10.5.
//
// Rate-limited, non-exclusive, and NON-ADAPTIVE -- it does not read the
// position. That access profile is what makes it a SOURCE rather than a
// JUDGE, and it is the whole of Prin. 10.9: a disbursement rule that read
// position quality would be an evaluation function held by the treasury,
// agents would be paid for agreeing with it, and the yield measurement
// would degenerate into measuring how well the ladder tracks the
// exchequer. That is the Rem. 5.5 leak arriving through the ACCOUNTS
// rather than through the floor, where it is much harder to see.
//
// CONSERVATION. Theta_W is fixed at the outset and never replenished, so
// the schedule REDISTRIBUTES a fixed endowment over time and does not
// mint: a large initial supply, meagrely distributed at first and
// increasingly as play goes on.  $\sigma + \kappa = \text{Theta}$  throughout.
//
// THE SCHEDULE IS KEYED TO CAPTURED MATERIAL, not to ply number. An
// exponential in ply diverges against a finite reserve, and one then
// harvests draws by exhaustion that are artefacts of the treasury rather
// than facts about the players. Captured material is bounded, monotone and
// self-normalising, and it says something better: the treasury pays out in
// proportion to how much of the position has been RESOLVED, so a simpler
// position funds a deeper search (Rem. 10.6).
//
// WHY THIS EXISTS. Harvest-only funding is PROCYCLICAL: the side that is
// behind has fewer agents and smaller stacks, so by Prin. 9.3 it calculates
// less far and falls further behind. That spiral is a CHOSEN property of a
// funding rule, not a structural feature of the architecture, and this
// contract is the correction. Note the convergence with Prop. 7.10:  $S$  falls
// as  $q$  rises, so per-agent budget rises SUPER-LINEARLY, and the endgame is
// where the society is simultaneously split-free by arithmetic and deepest
// by budget. Cor. 10.8: this player should be at its BEST in the endgame,
// the opposite of how weak engines usually fail, and therefore cheap to
// falsify (M11).
// =====

| contract Treasury(t, @captured, @reserve) = {
  for (@ids, ret <- t) {
    new q in {
      q!(Schedule(captured)) // non-decreasing in captured
    | for (@rate <- q) {
      new draw in {
        draw!(rate * Size(ids))
      | for (@total <- draw) {
        match reserve >= total {
          true => { Disburse!(ids, rate, *ret)
                    | Treasury!(*t, captured, reserve - total) }
          // Exhausted. The reserve is finite, and this is what the
          // end of a finite endowment looks like; it is not a
          // failure mode to be patched away.
          false => { ret!(0) | Treasury!(*t, captured, 0) }
        }
      }
    }
  }
}

// ASSAY-CONFIRMED DISBURSEMENT -- Def. 10.10, Prin. 10.11.

```



```

//
// The schedule above sets the FLOOR and removes the spiral; it pays per
// surviving agent regardless of how that agent is playing. This rule
// supplies the DIFFERENTIAL, and it is where selection lives. It reads only
// the confirm / refute / bottom trichotomy the assay ALREADY emits -- the
// same observable Revise uses to set its temperature -- so the judge is
// endogenous and no exogenous score is consulted.
//
// Under this rule a hypothesis that pays for itself BUYS ITS OWN DEPTH, and
// one that does not starves its holder. "Is the yield of a hypothesis worth
// its price" stops being a statistic computed after the games and becomes
// the mechanism by which the population changes. Build and measure it
// SEPARATELY from the schedule (Rem. 10.12): the first is a correction to a
// defect, the second is an experiment.
//
// NB the peek. Reading an agent's assay verdict must not consume it, for
// the same reason the settling peeks at occupancy: paying for an outcome
// must not destroy the record of it.
| contract Disburse(@ids, @rate, ret) = {
  match ids {
    [] => { ret!("done") }
    [id ...rest] => {
      for (@verdict <-- @[*player,"assay",id]) { // PEEK
        match verdict {
          "confirm" => { for (@sg <- @[*board,"m",id]) {
            @[*board,"m",id]!(sg + rate) } }
          _ => { Nil } // refuted or bottom: no draw
        }
      }
      | Disburse!(rest, rate, *ret)
    }
  }
}

// =====
// THE BOARD -- occupancy only. Note what is NOT here: a 'new' for the
// squares. Each square name is the quote of a term properly containing the
// board, so by the no-self-code theorem it cannot already occur in the
// board -- locally fresh, no registry consulted, decentralised by
// construction. And because the family is generated by a predicate on
// names, the whole namespace can be DESCRIBED rather than enumerated.
// =====

| contract Board(b, @placement) = { Cells!(*b, placement, 0) | Settle!(*b) }

| contract Cells(b, @placement, @s) = {
  match s < 64 {
    true => { @[*b,"occ",s]!(placement.nth(s))
      | Cells!(*b, placement, s + 1) }
    false => { Nil }
  }
}

// =====
// THE SETTLING -- Sec. 5.2 and Rem. 5.3.
//
// A ray walk PEEKS at occupancy ('<-') and never consumes it. That is what
// makes the scan perception rather than interaction: with a consuming
// receipt the settling would move the pieces by looking at them, and
// sixteen agents scanning concurrently would serialise on every square.
//
// Termination: each step consumes one square of a strictly decreasing

```

```

// remaining ray. Confluence: markers are deposited on the atk family, which
// nothing in the settling reads. That discharges the obligation <<K_m>>
// depends on, which in noughts was Lem. 8 and here is harder because the
// walk is recursive in occupancy.
// =====

```

```

| contract Ray(b, @from, @dir, @p, @c, @dist) = {
  match (dist > 0) and Onboard(from, dir) {
    true => {
      new here in {
        here!(Step(from, dir)) |
        for (@s <- here) {
          for (@v <- @[*b,"occ",s]) { // PEEK, never consume
            @[*b,"atk",s,c,p]!("bears", from))
            | match v {
              Nil => { Ray!(*b, s, dir, p, c, dist - 1) } // ray continues
              _ => { Nil } // blocked here
            }
          }
        }
      }
    }
    false => { Nil }
  }
}

```

```

// Every occupied square emits its piece's rays; then the kings emit theirs
// to deposit pins. Rem 5.4: this terminates and is confluent.

```

```

| contract Settle(b) = {
  new each in {
    Squares!(*b, *each)
    | for (@s <= each) {
      for (@v <- @[*b,"occ",s]) {
        match v {
          Nil => { Nil }
          _ => { Directions!(v, s, *b) } // per-type ray emission
        }
      }
    }
    | Pins!(*b)
  }
}

```

```

// A pin marker is deposited by walking OUT from a king through the first own
// piece to the first enemy slider on the same ray. Pin-awareness is
// therefore PERCEPTION: a piece does not reason about whether it is pinned,
// it reads whether a marker stands on its square. This is the note's best
// instance of read-versus-drive (Sec. 5.3).

```

```

| contract Pins(b) = {
  new k in {
    Kings!(*b, *k)
    | for (@(ksq, c) <= k) {
      new d in {
        Rays8!(*d)
        | for (@dir <= d) { PinWalk!(*b, ksq, dir, c, Nil) }
      }
    }
  }
}

```

```

// =====
// A PLY IS A JOIN -- Sec. 6, Def. 6.1 and Prop. 6.2.

```

```

//
// One N-ary join over the occupancy cells the move touches, guarded on the
// constraint family, re-emitting every consumed cell with its post-move
// payload. A failed guard leaves the messages in the tuple space, which is
// the platform's own semantics for a failed 'where', so a refusal is a
// no-op and not a rollback.
//
// ARITY IS THE MOVE TAXONOMY:
// 2 quiet move (*p, Nil) |-> (Nil, *p)
// 2 capture (*p, *q) |-> (Nil, *p), q terminated
// 3 en passant (*p, Nil, *q) |-> (Nil, *p, Nil), q terminated
// 4 castling (*k, Nil, Nil, *r) |-> (Nil, *k, *r, Nil)
// 2 promotion (*p, .) |-> (Nil, *p'), p retired
//
// This is what keeps the redex positions of the ply rule in bijection with
// the legal moves, and hence what keeps <K_m> a GENERATED modality rather
// than a derived abbreviation (Cor. 6.3).
// =====

| contract Ply(b, @m, ret) = {
  match Kind(m) {

    // --- quiet move: arity 2 -----
    "quiet" => {
      for ( @vf <- @[*b, "occ", From(m)] & @vt <- @[*b, "occ", To(m)]
        where Legal(m, vf, vt) ) { //! ideal
        @[*b, "occ", From(m)]! (Nil)
        | @[*b, "occ", To(m)]! (vf)
        | Settle!(*b) | ret!(("played", m))
      }
    }

    // --- capture: arity 2, and the occupant is EATEN -----
    // Harvest routes the captured agent's metabolic name to the capturer.
    // The arena mints nothing; predation is legal here for precisely one
    // reason, and it is that somebody took a piece.
    "capture" => {
      for ( @vf <- @[*b, "occ", From(m)] & @vt <- @[*b, "occ", To(m)]
        where Legal(m, vf, vt) ) { //! ideal
        @[*b, "occ", From(m)]! (Nil)
        | @[*b, "occ", To(m)]! (vf)
        | Harvest!(Stack(vt), Stack(vf), *display)
        | Settle!(*b) | ret!(("played", m))
      }
    }

    // --- castling: arity 4, all-or-nothing -----
    // The same shape as the four-cell SWAP of the cost-accounting demo:
    // the guard decides BEFORE any leg fires, so no interleaving can
    // leave the rook moved and the king not.
    "castle" => {
      for ( @vk <- @[*b, "occ", KFrom(m)] & @v1 <- @[*b, "occ", KTo(m)]
        & @v2 <- @[*b, "occ", RTo(m)] & @vr <- @[*b, "occ", RFrom(m)]
        where CastleLegal(m, vk, v1, v2, vr) ) { //! ideal
        @[*b, "occ", KFrom(m)]! (Nil) | @[*b, "occ", KTo(m)]! (vk)
        | @[*b, "occ", RFrom(m)]! (Nil) | @[*b, "occ", RTo(m)]! (vr)
        | Settle!(*b) | ret!(("played", m))
      }
    }

    // en passant (arity 3) and promotion follow the same pattern.
    _ => { ret!(("refused", m)) }
  }
}

```

```

    }
  }

// Conservation: nothing minted, the loser's stack MOVED into the winner's.
| contract Harvest(mLost, mWon, ret) = {
  for (@a <- mLost & @c <- mWon) { mWon!(a + c) | ret!("ate", a) }
}

// =====
// THE FRANCHISE -- Sec. 7.4. One LINEAR advocacy token per live piece per
// round, into a located stack whose channel is manufactured from the
// piece's identity. This is the polity listing's pre-election phase
// verbatim: a persistent claim on the roll metered to one linear token per
// epoch. It is a SOURCE keyed to identity, not a mint.
//
// It is also what makes Thm. 7.6 arithmetic rather than statistical.
// =====

| contract Franchise(@ids, @r) = {
  match ids {
    [] => { Nil }
    [id ...rest] => {
      @[*player,"V",id,r]!("vote", id, r)
      | Franchise!(rest, r)
    }
  }
}

// =====
// A PIECE -- Sec. 12.3, steps 4-6 of the turn protocol.
//
// A piece enumerates ITS OWN moves from ITS OWN square. Nobody computes the
// union of candidates; the union is what the advocacy namespace IS.
// Evaluation is an ASSAY: the rung sits in guard position, so evaluating is
// testing and confirmation is the rendezvous. By Prop. 17 of the scientist
// note the number of candidates a piece can afford to evaluate is its stack
// divided by the price of one check, so a poor piece does not survey its
// own branching -- and since DEPTH is bought from the same stack, a poor
// piece does not look far either.
// =====

| contract Piece(b, @id, @rung, @r, m) = {
  new cand, best in {
    Candidates!(*b, id, *cand)
    | for (@c <- cand where Sat(c, rung, id)) { /// ideal
      best!(c)
    }
    | for (@c <- best) { Advocate!(id, c, r) }
  }
}

// Spending the token IS the advocacy. A candidate move c is a term, so @c is
// a name; the advocacy channel is manufactured from it. An OPEN in the
// coalition sense is the set of pieces that have advocated for a given move
// this round -- described by a name predicate, never enumerated.
| contract Advocate(@id, @c, @r) = {
  for (@tok <- @[*player,"V",id,r]) { /// LINEAR: spent once
    @[*player,"adv",@c]!(id)
  }
}

// =====

```

```

// THE COMMIT GATE -- Sec. 7.5, Def. 7.5.
//
// QUORUM IS A JOIN ARITY. The gate is a k-ary join on one advocacy channel;
// it fires exactly when k distinct advocacy sends stand there, and it
// CONSUMES them. Shown for k = 3; the general case is generated.
//
// THM 7.6. Two gates on distinct candidates consume 2k tokens drawn from a
// pool of S linear tokens, so two quorums require S >= 2k. With S = 16 and
// k >= 9 a SPLIT IS STRUCTURALLY IMPOSSIBLE, for any policies whatever, and
// no cross-inhibition machinery is needed to secure it. Safety is free;
// liveness is not, and that is what Round is for.
//
// THE PRICE OF THE GOVERNANCE LAYER. An engine takes the MAXIMUM over its
// candidates; a quorum takes a PLURALITY above a threshold. So the society
// can beat its average member and will struggle to beat its best one, and
// a piece holding the right move loses to eight pieces holding a worse one.
// That is the architecture, not a defect of it -- a maximum over sixteen
// agents needs an evaluator none of them can afford to be. But do NOT read
// it as an established structural weakness. Against the commit gap runs a
// COVERAGE GAIN the same architecture supplies: sixteen agents at
// heterogeneous depths over different candidate sets examine a wider tree
// than any one of them could (Obs. 9.17 read as a benefit rather than an
// overhead). Which dominates is not known and is not obvious in either
// direction; M10 nets the two, and Q12 asks what can be bought back
// without reintroducing a chairman.
// =====

| contract Gate(@c, @k, ret) = {
  for ( @a1 <- @[*player,"adv",@c]
    & @a2 <- @[*player,"adv",@c]
    & @a3 <- @[*player,"adv",@c] ) {
    @[*player,"commit"]!(@c) | ret!(("quorum", @c, [a1,a2,a3]))
  }
}

// =====
// THE ROUND -- Sec. 7.6, Def. 7.8 and Def. 7.9.
//
// Unconsumed advocacy is LEFT STANDING and is readable, so a piece may
// re-advocate having PERCEIVED where the rest of the society stood. That is
// the bees reading dance strength at the cluster: cheap perceptual price for
// the reversible decision, expensive interaction price for the irreversible
// one. Cross-inhibition is deliberately absent -- it is one more mechanism
// and the swarm measurement recorded that it suppresses CORRECT advocacy
// too. Deadlock is broken by re-advocacy and, failing that, by the clock.
// =====

| contract Round(b, @ids, @r, @maxR, ret) = {
  new fired in {
    Franchise!(ids, r)
    | Pieces!(*b, ids, r)
    | Gates!(ids, r, *fired)
    | for (@res <- fired) { ret!(res) }
    | for (@t <- Timeout(r)) { /// ideal
      match r < maxR {
        true => { Round!(*b, ids, r + 1, maxR, *ret) }
        // Resolver of last resort: the CLOCK, not a piece. It holds no
        // evaluation function and no preferences; its firing rate is a
        // measurement (M4), because a society that reaches it often is a
        // society whose quorum parameters are wrong.
        false => { MostAdvocated!(ids, r, *ret) }
      }
    }
  }
}

```

```

    }
  }
}

// =====
// THE LADDER -- Sec. 12.4. Cumulative, highest priority first. The rung
// index is the modal depth of the conjunct it adds, hence the ultrametric
// stage:  $d(\text{Phi}_r, \text{Phi}_{\{r+1\}}) = 2^{-r}$ . Prices are  $\kappa(d) = \beta^d$  per
// candidate examined -- and for the modal rungs that is DERIVED, not
// stipulated: it is the cost of instantiating a shadow board and driving
// it d plies (Prop. 9.8). The one free parameter is beta.
//
// Note where the alternation is. Rungs 1, 2 and 5 have no modality at all:
// capturing, not hanging, and centralising are PERCEPTION, reading the
// markers the last settling deposited. The one genuinely alternating rung
// is 4, a box under a diamond, which by Prop. 6 of the scientist note has
// no finite confirming witness. In noughts that rung was also, measurably,
// the worst purchase on the ladder. Whether that repeats in chess is M2 --
// but rung 4 in noughts is a two-ply search on a 5478-position game and
// rung 4 here is a two-ply search on CHESS, so the ratio must be taken at
// MATCHED DEPTH or it compares different instruments.
//
// The rung fixes the SHAPE of what an agent asks; its STACK fixes how far.
// Depth is therefore endogenous and per-agent (Prin. 9.3): an agent that
// has eaten calculates further than one that has not. NB this is a CHOSEN
// dial, not a structural property -- depth could be funded from a per-turn
// allowance instead, and Sec. 14 says so.
//
// WHAT THIS LADDER IS. It is an evaluation function, and a thin one. That
// is a SHORT LADDER, not a weak logic: the generated logic has nu, graded
// separation over described namespaces, the name predicate and
// redex-indexed modalities, and under reflection reaches further again. An
// earlier draft called the coarseness forced on the grounds that a rung
// must be a formula of the logic; that confused expressibility with length.
//
// WHAT ACTUALLY GOVERNS IS PRICE, and price turns on what the world has
// DEPOSITED. Rungs 5a and 5b below are the demonstration: king safety and
// pawn structure are classically computed per evaluated position, and
// deposited by the settling walk they are depth-ONE reads. This is
// read-versus-drive (Sec. 5.2) applied to EVALUATION rather than to
// legality, which is why M7 is a larger experiment than it first looks --
// the question is not merely who pays for the attack map, but how much of
// an evaluation function the world can be made to compute once for all.
//
// WHAT IS STILL OPEN IS FINDABILITY (Q11): not whether the logic can say a
// good evaluation, nor whether an agent can afford one, but whether the
// revision process can FIND one. Confinement says an incremental walk never
// leaves its initial ball, and choosing the SHAPE of an evaluation is a
// distance- $2^{-1}$  move, so no individual gets there -- which leaves the
// population and its crossover operator as the only route.
// =====

//! ideal
// Emp(s) := out(occ(s))@Nil
// Bears(p,s) := exists c. out(atk(s,c,p)) TOP
// Fork(p) := Threat(p) |_0^{N_atk} Threat(p)
// Bearing(s) :=  $\langle E \rangle_{\{N_{pl}\}}^V \lambda p. \text{Bears}(p,s) (x) \text{Available}(p)$ 
// KingSafe(c) :=  $\langle E \rangle_{\{N_{atk}\}}^V \lambda p. \text{exists } s. \text{kz}(c,s) \text{ and}$ 
// Bears(p,s) and Colour(p, not c)
// Worsens(m,Pawns) :=  $\langle \langle K \ m \rangle \rangle \text{PawnDebt} > \text{PawnDebt}$ 
//  $\langle \langle K \ m \rangle \rangle \text{phi}$  := the ply join at m fires, the board settles, phi holds
| contract Ladder(@r, ret) = {

```

```

match r {
  0 => { ret!( @ { TOP } ) }
  1 => { ret!( @ { exists m. Capture(m) and not Defended(to(m)) } ) }
  2 => { ret!( @ { exists m. Capture(m) and not Defended(to(m))
    or not Hangs(m) } ) }
  3 => { ret!( @ { exists m. Capture(m) and not Defended(to(m))
    or not Hangs(m)
    or <<K m>> Fork(me) } ) }
  4 => { ret!( @ { exists m. Capture(m) and not Defended(to(m))
    or not Hangs(m)
    or <<K m>> Fork(me)
    or forall m2. <<K m>>[K m2] not Fork(them) } ) }
  5 => { ret!( @ { exists m. Capture(m) and not Defended(to(m))
    or not Hangs(m)
    or <<K m>> Fork(me)
    or forall m2. <<K m>>[K m2] not Fork(them)
    or Central(to(m)) or Develops(m) } ) }
  // 5a, 5b: depth-ONE reads over deposited structure, Def. 12.3 and
  // 12.4. kz(c,s) is one extra walk in the settling, over a
  // neighbourhood of bounded size, in the pass that already walks every
  // ray; pwn(c,f,kind) is a multiplicity condition on one file and
  // negations over the adjacent ones. Neither needs a search and
  // neither needs more logic than the ladder already used.
  51 => { ret!( @ { exists m. Capture(m) and not Defended(to(m))
    or not Hangs(m)
    or <<K m>> Fork(me)
    or forall m2. <<K m>>[K m2] not Fork(them)
    or Central(to(m)) or Develops(m)
    or <<K m>> KingSafe(me) } ) }
  52 => { ret!( @ { exists m. Capture(m) and not Defended(to(m))
    or not Hangs(m)
    or <<K m>> Fork(me)
    or forall m2. <<K m>>[K m2] not Fork(them)
    or Central(to(m)) or Develops(m)
    or <<K m>> KingSafe(me)
    or not Worsens(m, Pawns) } ) }
  // The FORCING rung, Def. 9.10. <E> does not mean "this coalition has
  // the resources"; it means the coalition can FORCE the goal whatever
  // the complement does -- and the complement is the OPPONENT'S pieces.
  // So the opponent enters as the complement of a coalition rather than
  // as a second search agent to be simulated, and rung 4 is this rung at
  // fixed-point depth one with a weaker target (Prop. 9.11).
  // Attained under (x) = min, so Check returns the WITNESS: at depth 0
  // the coalition, at depth d the LINE.
  6 => { ret!( @ { exists m. exists s. <<K m>>
    <E>_{N_pl}^V nu X. (Mate(s) (+) <K> X) } ) }
  // chi is listed for completeness. Unlike noughts it is NOT exact at any
  // finite unfolding, and by Rem. 3.1 that is exactly the condition under
  // which the ecology does not die of its own perfection.
  7 => { ret!( @ { nu X. forall m. [K m] (not Lost
    and exists m2. <<K m2>> X) } ) }
}
}

// Revision walks the lattice. The move has the two parameters the metric
// grades: a LOCUS and a DIRECTION. Refutation rate is the temperature, and it
// is an observable the assays already emit, not a schedule imposed from
// outside. NB no case touches the piece's ADVOCACY STYLE: that is the
// stage-1 approximant, a move of distance 2^-1, and an individual can
// neither reach it by deepening nor afford it if it could. It changes across
// generations, by crossover, and that is where a model-backed mutation
// operator earns its place (Sec. 11.2).

```

```

| contract Revise(@rung, @refutes, @confirms, ret) = {
  match refutes * 4 > confirms {
    true => { ret!(rung + 1) }
    false => { ret!(rung) }
  }
}

// =====
// THE CHECKER -- Sec. 12.6. An instrument the player HOLDS, sited in the
// world so that sixteen agents share one cache. It receives only on the
// deposited families and NEVER on occ: otherwise the pride would move the
// zebra by looking at it. Verdicts are Kleene {f < b < t} with b carrying
// both readings -- the environment is stuck, or the observer could not
// afford to decide -- and the caller must not be able to tell which.
// =====

| contract Check(@formula, @posHandle, @algebra, @depth, payer, ret) = {
  new r in {
    // Price is  $\kappa(d) = \beta^d$  SHADOW PLIES, not a stipulated
    // constant: Prop. 9.8.  $\beta$  is measured, not set, and it is exactly
    // what a pruning discipline reduces.
    Live!(*payer, Price(formula, depth), *r)
    | for (_ <- r) {
      match Modal(formula) {
        // Depth 0: read the deposited families. PERCEPTION.
        false => { Evaluate!(formula, posHandle, algebra, *ret) }
        // Depth d: quote the deposit, manufacture a private board, and
        // drive it. REFLECTION. The witness at depth d is a PRINCIPAL
        // VARIATION (Prop. 9.13); at depth 0 it is a coalition.
        true => { Shadow!(posHandle, depth, 0, formula, algebra, *ret) }
      }
    }
    // If Live cannot fund the query the piece receives nothing, which the
    // protocol reads as verdict "b" -- here in its search-shaped form, "I
    // ran out of depth", which is indistinguishable from "the position is
    // genuinely unresolved". Two bottoms, one value, by design.
  }
}

// =====
// THE SHADOW BOARD -- Sec. 9.2, Def. 9.5 and Req. 9.6.
//
// REQ 9.6. The checker may not move THE board: on the live board it PEEKS
// ('<<-') and never consumes, on any family. It may receive freely on the
// families of a board it has MANUFACTURED. A shadow name is the quote of a
// term properly containing the checker, so by the no-self-code theorem it
// is fresh in the live board and the two families CANNOT collide. The
// separation is structural, not disciplinary -- the same argument that put
// attack markers on their own family in Sec. 5.1, one level up.
//
// Each ply driven here is one metered step, so the recursion IS the search
// tree and Prop. 9.8 reads off its price. The alternation is fixed by the
// formula; the ORDER in which candidates are examined is not, which is why
// pruning is a scheduler discipline and not a different formula (Obs 9.12).
// =====

| contract Shadow(@H, @d, @i, @formula, @algebra, ret) = {
  new sb in {
    sb!(@[*Check, H, d, i]) // manufactured, hence fresh
    | for (@b2 <- sb) {
      match d {
        0 => { Evaluate!(formula, b2, algebra, *ret) }
      }
    }
  }
}

```



```

_ => {
  new cand, acc in {
    Board!(b2, Placement(H)) // instantiate the copy
    | ShadowMoves!(b2, *cand)
    | for (@m <= cand) {
      new r2 in {
        Ply!(b2, m, *r2) // the COPY may be moved
        | for (@o <- r2) {
          Shadow!(Deposit(b2), d - 1, i + 1,
            Peel(formula), algebra, *acc)
        }
      }
    }
    // Over an attained algebra (tropical, Viterbi) the (+) is
    // realised by a definable selection, so Combine returns
    // the WITNESS LINE and not merely its value. A pruning
    // discipline is sound exactly when it leaves that witness
    // alone (Cor. 9.14) -- so a changed principal variation
    // under pruning is a soundness bug, not noise.
    | Combine!(*acc, algebra, *ret)
  }
}

}

}

}

// =====
// THE ARENA -- the rules, holding no stack of its own. It mints nothing;
// it ROUTES. An exogenous reward would need a referee with a mint, and a
// referee with a mint destroys conservation.
//
// It is also where the one unavoidable universal lives. Check, pin and
// discovered check constrain the legality of EVERY piece's move; no piece
// can establish that locally and no quorum can establish it either, because
// an illegal move is not a move and a vote does not make it one. So the
// world refuses, and the player's own hypothesis language stays existential
// and finitely confirmable.
// =====

| contract Turn(b, @ids, @maxR, @who, over) = {
  new chosen in {
    Round!(*b, ids, 1, maxR, *chosen)
    | for (@res <- chosen) {
      new played in {
        Ply!(*b, Move(res), *played)
        | for (@out <- played) {
          match out {
            ("played", m) => {
              Adjudicate!(*b, *over, who)
              | Turn!(*b, Live_ids(ids), maxR, 3 - who, *over)
            }
            // A refused guard left the messages in the tuple space, so
            // the position is untouched and the clock simply retries.
            ("refused", m) => { Turn!(*b, ids, maxR, who, *over) }
          }
        }
      }
    }
  }
}

}

}

}

```

```

//! ideal -- mate, stalemate and the draw conditions are modal formulae over
// the deposited families, not scans of a list. The fifty-move and repetition
// rules are the only genuinely HISTORICAL facts in chess and nothing here
// uses the history monad. That is open problem Q4, and it is now the
// largest item in the design: a SEARCHED fact -- this deposit was driven
// to depth d and returned v -- is a configuration paired with the word of
// events that produced it, which is the history monad's own object. The
// transposition table is its ecological reading, and the duplication it
// removes grows with beta^d rather than with S (Sec. 9.5).
| contract Adjudicate(b, over, @who) = {
  new r in {
    @[*b,"state"]!(*r)
    | for (@pos <- r where pos |= @{ Mate(3 - who) }) {
      over!("won", who))
    }
    | for (@pos <- r where pos |= @{ Stalemate(3 - who) }) {
      over!("draw")
    }
  }
}

| Boot!(Nil)
}

```

B The formula catalogue

Structural predicates, all over deposited families, all at perception prices:

$$\begin{aligned}
\text{Occ}(s, p) &:= \text{out}(\text{occ}(s)) @(*p) \\
\text{Emp}(s) &:= \text{out}(\text{occ}(s)) @\mathbf{0} \\
\text{Bears}(p, s) &:= \exists c. \text{out}(\text{atk}(s, c, p)) \top \\
\text{Attacked}(s, c) &:= \exists p. \text{Bears}(p, s) \wedge \text{Colour}(p, c) \\
\text{Defended}(s) &:= \text{Attacked}(s, me) \\
\text{Pinned}(p) &:= \exists s. \text{Occ}(s, p) \wedge \text{out}(\text{pin}(s)) \top \\
\text{Available}(p) &:= \neg \text{Pinned}(p) \wedge \neg \text{Committed}(p) \\
\text{InCheck}(c) &:= \text{out}(\text{chk}(c)) \top
\end{aligned}$$

Derived formulae:

$$\begin{aligned}
\text{Threat}(p) &:= \exists s. \text{Bears}(p, s) \wedge \text{Valuable}(s) \\
\text{Fork}(p) &:= \text{Threat}(p) \mid_0^{N_{\text{atk}}} \text{Threat}(p) \\
\text{Hangs}(m) &:= \text{Attacked}(\text{to}(m), \text{them}) \wedge \neg \text{Defended}(\text{to}(m)) \\
\text{Bearing}(s) &:= \langle E \rangle_{N_{pl}}^V \lambda p. \text{Bears}(p, s) \otimes \text{Available}(p) \\
\text{Forces}(s) &:= \langle E \rangle_{N_{pl}}^V \nu X. (\lceil \text{Mate}(s) \rceil \oplus \langle K \rangle X) \\
\text{KingSafe}(c) &:= \langle E \rangle_{N_{\text{atk}}}^V \lambda p. (\exists s. \text{kz}(c, s) \wedge \text{Bears}(p, s) \wedge \text{Colour}(p, \bar{c})) \\
\text{Worsens}(m, \text{Pawns}) &:= \langle \langle K_m \rangle \rangle \text{PawnDebt} > \text{PawnDebt} \\
\text{Coherent} &:= \nu X. \langle E \rangle_{N_{pl}}^V (\lceil \beta \rceil \otimes \langle K \rangle X)
\end{aligned}$$

Coherent is the society's analogue of herd coherence: the coalition keeps agreeing as the population steps. It is not on the ladder because it is a ν over the run tree and therefore

a depth engine, priced accordingly; it is listed because it is the formula an *observer* of the society would use, and the society is going to be observed.